

What is Chemistry, and What Can Chemistry Do for You?

You may wonder why

- Why the boiling water bubbles and produces steam while the teakettle retains its original shape?
- How do the tea leaves change the color of the water while the teabag remains as full and plump as ever?
- Why does sugar make your tea sweet, and why is the tea itself bitter?
- How gasoline burns and drives vehicles down the road?
- How does it pollute the air we breathe?
- Why some books that are hundreds of years old are still in good shape while other books that are only 50 years old have pages that are brown, brittle, and crumbling.
- Can the books with damaged pages be saved?

Chemists can answer all these questions
and
Others like them



freepresspicture.com



Dreamstime.com



Wikipedia.org



Periodic Patterns in the Main-Group Elements

Group

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

1	1																	
2	3	4												5	6	7	8	9
3	11	12												13	14	15	16	17
4	19	20	21	22	23	24	25	26	27	28	29	30		31	32	33	34	35
5	37	38	39	40	41	42	43	44	45	46	47	48		49	50	51	52	53
6	55	56	71	72	73	74	75	76	77	78	79	80		81	82	83	84	85
7	87	88	103	104	105	106	107	108	109	110	111	112		113	114	115	116	117

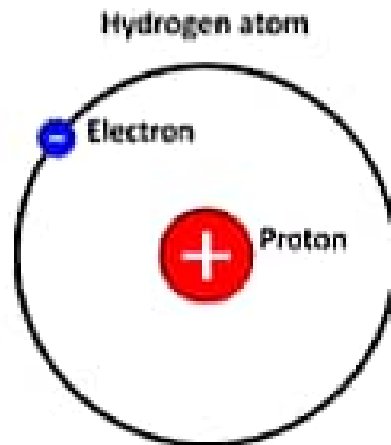
Dr. Ibrahim Alkhatib

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- 14.2 Trends Across the Periodic Table:
The Period 2 Elements
- 14.3 Group 1A(1): The Alkali Metals
- 14.4 Group 2A(2): The Alkaline Earth Metals
- 14.5 Group 3A(13): The Boron Family
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- 14.6 Group 4A(14): The Carbon Family
Highlights of Carbon Chemistry
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- 14.7 Group 5A(15): The Nitrogen Family
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- 14.8 Group 6A(16): The Oxygen Family
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- 14.10 Group 8A(18): The Noble Gases

14.1 HYDROGEN, THE SIMPLEST ATOM



- Nucleus with a single positive charge
- Surrounded by a single electron

Importance of hydrogen



- Despite this simple structure, or perhaps because of it, hydrogen may be the most important element of all.
- In the Sun, hydrogen (H) nuclei combine to form helium (He) nuclei in a process that provides nearly all Earth's energy.
- About 90% of all the atoms in the universe are H atoms, making it the most abundant element by far.
- On Earth, only tiny amounts of the free element occur naturally (as H_2), but hydrogen is abundant in combination with oxygen in water.

Properties of H_2 gas

- Colorless
- Odorless
- Very low melting ($-259^\circ C$)
- Boiling points ($-253^\circ C$).

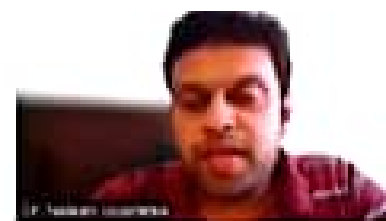


Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Some think it fits in Group 1A(1) because of its single valence electron and common 1 oxidation state, **However**,
 - Unlike the alkali metals, hydrogen shares its electron with nonmetals rather than losing it to them.
 - Like a nonmetal, it has a relatively high ionization energy (IE 1311 kJ/mol) much higher than that of lithium (IE 520 kJ/mol), whose IE is the highest in Group 1A
 - Much higher electronegativity (EN of H 2.1; EN of Li 1.0).

	1A	2A	3A	4A					
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)	
1	H			H			H		
2									
3									
4									
5									
6									
7									



Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Others think hydrogen fits in Group 7A(17) because like the halogens it forms diatomic molecules and fills its outer (valence) level either by electron sharing or by gaining one electron from a metal to form a monatomic anion, the hydride ion (H^- ; O.N. = -1).
- However, hydrogen has a lower electronegativity ($EN = 2.1$) than any of the halogens (ENs range from 2.2 to 4.0)
- Lacks their three valence electron pairs.
- H^+ ion is rare and reactive, whereas halide ions are common and stable.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								



Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Still others think hydrogen fits best in Group 4A(14) because of a half-filled valence level and similarities in ionization energy, electron affinity, electronegativity, and bond energies.

Hydrogen's unique behavior is attributable to its tiny size.

- Hydrogen has a high IE because its electron is very close to the nucleus, with no inner electrons to shield it from the positive charge.
- It has a low EN (for a nonmetal) because it has only one proton to attract bonding electrons.
- In this chapter, hydrogen will appear in either Group 1A(1) or 7A(17) depending on the property being considered.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Highlights of Hydrogen Chemistry

Ionic (Saltlike) Hydrides

- With very reactive metals, such as those in Group 1A(1) (Li, Na, K) and the larger members of Group 2A(2) (Ca, Sr, and Ba), hydrogen forms *saltlike hydrides*-white, crystalline solids composed of the metal cation and the hydride ion:

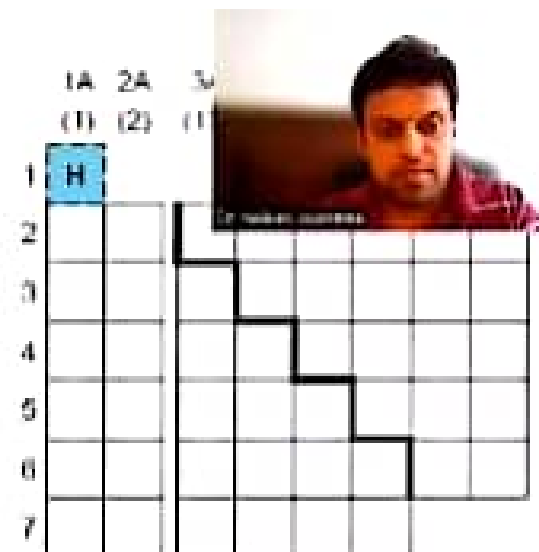


KH

- In water, H^- is a strong base that pulls H^+ from surrounding H_2O molecules to form H_2 and OH^- :



- The hydride ion is also a powerful reducing agent; for example, it reduces Ti(IV) to the free metal:



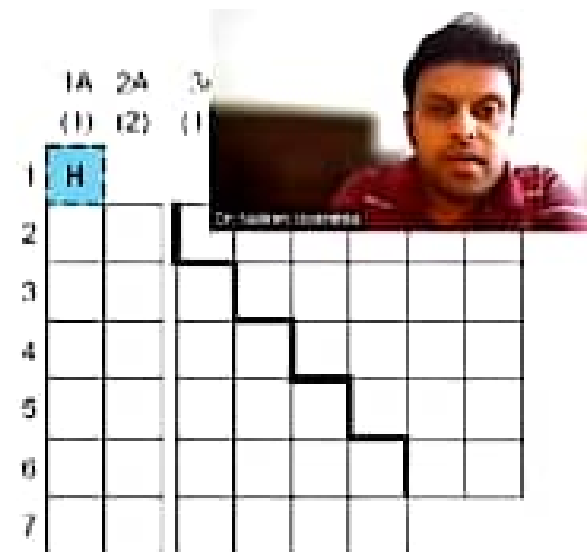
Highlights of Hydrogen Chemistry

Covalent (Molecular) Hydrides

- Hydrogen reacts with nonmetals to form many covalent hydrides, such as CH_4 , NH_3 , H_2O and HF .
- In most covalent hydrides, hydrogen has an oxidation number of +1 because the other nonmetal has a higher electronegativity than hydrogen.
- Conditions for preparing the covalent hydrides depend on the reactivity of the other nonmetal.
 - For example, with stable, triple-bonded N_2 , hydrogen reacts at high temperatures ($\sim 400^\circ\text{C}$) and pressures ($\sim 250\text{ atm}$), and the reaction needs a catalyst to proceed at any practical speed:



- hydrogen combines rapidly with reactive, single-bonded F_2 , even at extremely low temperatures (-196°C):

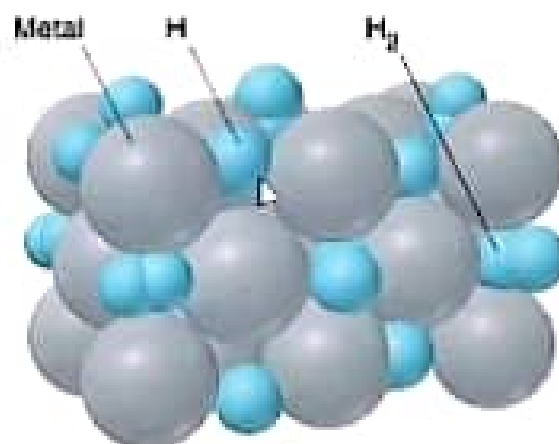


Highlights of Hydrogen Chemistry



Metallic (Interstitial) Hydrides

- Many transition elements form metallic (interstitial) hydrides, in which H_2 molecules (and H atoms) occupy the holes in the metal's crystal structure. Thus, such hydrides are not compounds but gas-solid solutions.



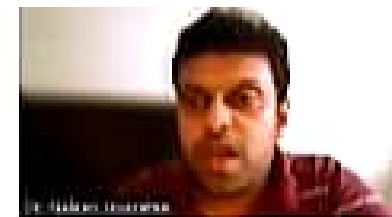
Interstitial Hydrides

- Also, unlike ionic and covalent hydrides, interstitial hydrides, such as TiH_x , typically lack a single, stoichiometric formula because the metal can incorporate variable amounts of hydrogen, depending on the pressure and temperature of the gas.



14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS

- Electrons fill the one ns and the three np orbitals according to Pauli's exclusion principle and Hund's rule.
- As a result of increasing nuclear charge and the addition of electrons to orbitals of the same energy level (same n value), atomic size generally decreases, whereas first ionization energy and electronegativity generally increase (see bar graphs on p. 557).
- Metallic character decreases with increasing nuclear charge as period members change from metals to metalloids to nonmetals.
- Reactivity is highest at the left and right ends of the period, except for the inert noble gas.
- Bonding between atoms of an element changes from metallic, to covalent in networks, to covalent in individual molecules, to none (noble gases exist as separate atoms). As expected, physical properties, such as melting point, change abruptly at the network/molecule boundary, which occurs in Period 2 between carbon (solid) and nitrogen (gas).
- Bonding between each element and an active nonmetal changes from ionic, to polar covalent, to covalent. Bonding between each element and an active metal changes from metallic to polar covalent to ionic.



14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS

- The acid-base behavior of the common element oxide in water changes from basic to amphoteric to acidic as the bond between the element and oxygen becomes more covalent.
- Reducing strength decreases through the metals, and oxidizing strength increases through the nonmetals. In Period 2,
 - Common oxidation numbers (o.N.s) equal the A-group number for Li and Be
 - A-group number minus eight for O and F
 - Boron has several o.N.s
 - Ne has none, and
 - C and N show all possible o.N.s for their groups

Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

Group: Element/At. No.:	1A(1) Lithium (Li) $Z = 3$	2A(2) Beryllium (Be) $Z = 4$	3A(13) Boron (B) $Z = 5$	4A(14) Carbon (C) $Z = 6$
Atomic Properties				
Condensed electron configuration; partial orbital diagram	$[\text{He}] 2s^1$ 	$[\text{He}] 2s^2$ 	$[\text{He}] 2s^2 2p^1$ 	$[\text{He}] 2s^2 2p^2$ 
Physical Properties				
Appearance				
Metallic character	Metal	Metal	Metalloid	Nonmetal
Hardness	Soft	Hard	Very hard	Graphite: soft Diamond: extremely hard
Melting point/ boiling point	Low mp for a metal	High mp	Extremely high mp	Extremely high mp

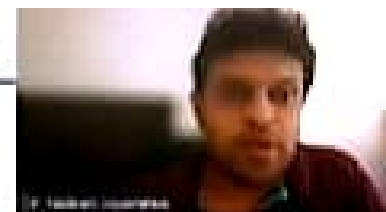


Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

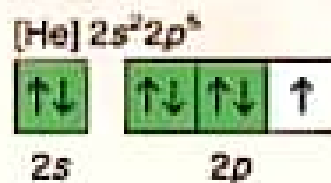
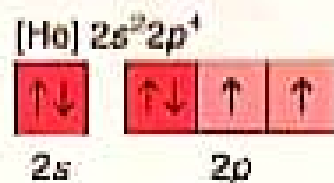
Group: Element/At. No.:	1A(1) Lithium (Li) $Z = 3$	2A(2) Beryllium (Be) $Z = 4$	3A(13) Boron (B) $Z = 5$	4A(14) Carbon (C) $Z = 6$
Chemical Properties				
General reactivity	Reactive	Low reactivity at room temperature	Low reactivity at room temperature	Low reactivity at room temperature; graphite more reactive
Bonding among atoms of element	Metallic	Metallic	Network covalent	Network covalent
Bonding with nonmetals	Ionic	Polar covalent	Polar covalent	Covalent (π bonds common)
Bonding with metals	Metallic	Metallic	Polar covalent	Polar covalent
Acid-base behavior of common oxide	Strongly basic	Amphoteric	Very weakly acidic	Very weakly acidic
Redox behavior (O.N.)	Strong reducing agent (+1)	Moderately strong reducing agent (+2)	Complex hydrides; good reducing agents (+3, -3)	Every oxidation state from +4 to -4
Relevance/Uses of Element and Compounds				
	Li soaps for auto grease; thermo-nuclear bombs; high-voltage, low-weight batteries; treatment of bipolar disorders (Li_2CO_3)	Rocket nose cones; alloys for springs and gears; nuclear reactor parts; x-ray tubes	Cleaning agent (borax); eyewash, antiseptic (boric acid); armor (B_4C); borosilicate glass; plant nutrient	Graphite: lubricant, structural fiber Diamond: jewelry, cutting tools, protective films Limestone (CaCO_3) Organic compounds: drugs, fuels, textiles, etc.

5A(15)
Nitrogen (N) $Z = 7$

6A(16)
Oxygen (O) $Z = 8$

7A(17)
Fluorine (F) $Z = 9$

8A(18)
Neon (Ne) $Z = 10$



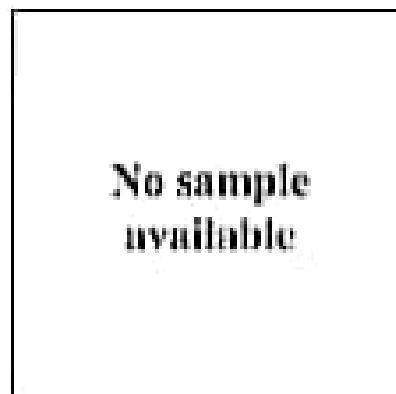
Nonmetal

Very low mp
and bp



Nonmetal

Very low mp
and bp



Nonmetal

Very low mp
and bp

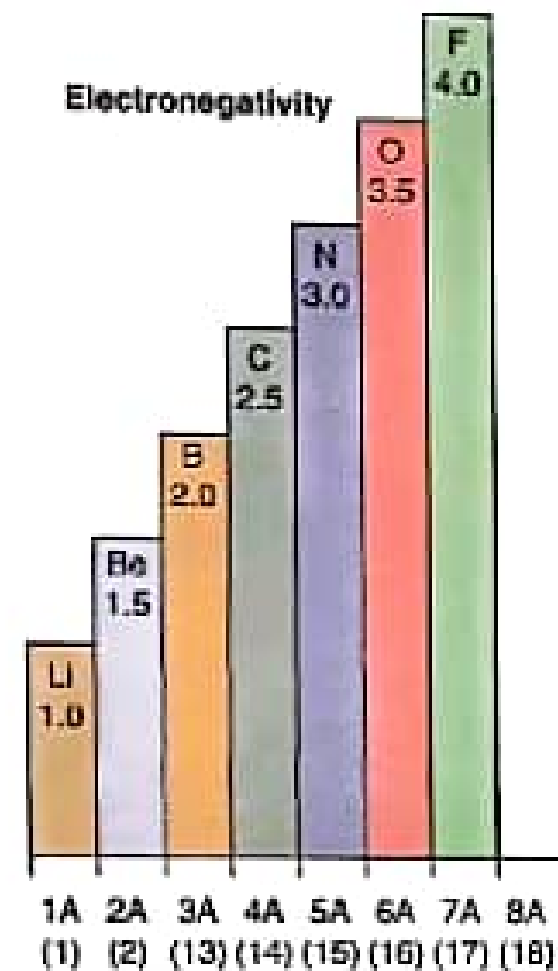
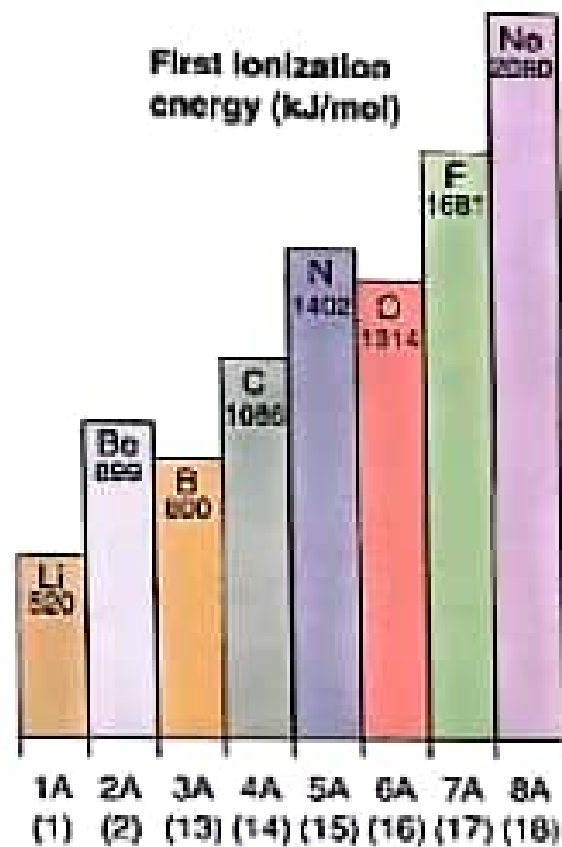
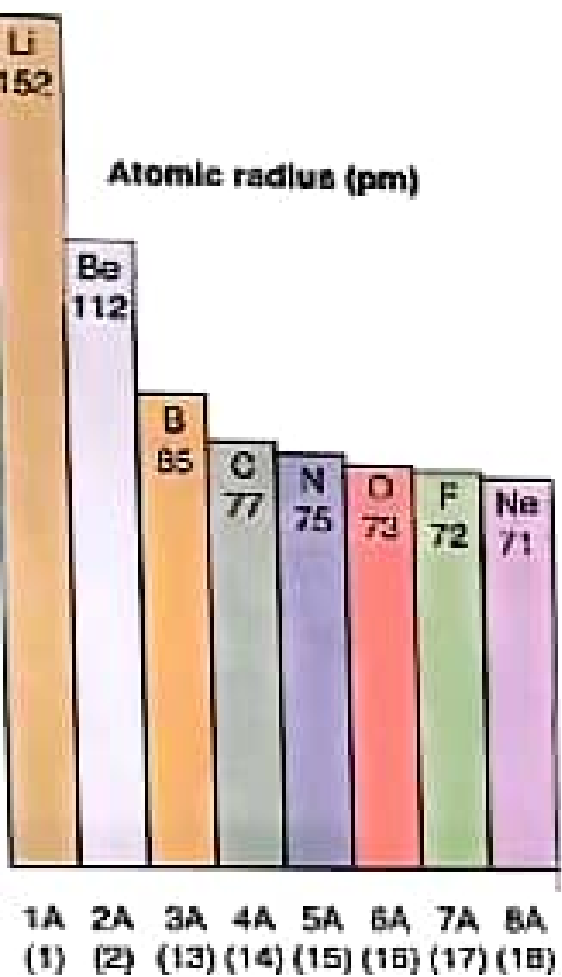


Nonmetal

Extremely low mp
and bp

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	5A(15) Nitrogen (N) $Z = 7$	6A(16) Oxygen (O) $Z = 8$	7A(17) Fluorine (F) $Z = 9$	8A(18) Neon (Ne) $Z = 10$
Chemical Properties				
General reactivity	Inactive at room temperature	Very reactive	Extremely reactive	Chemically inert
Bonding among atoms of element	Covalent N_2 molecules	Covalent O_2 (or O_3) molecules	Covalent F_2 molecules	None; separate atoms
Bonding with nonmetals	Covalent (π bonds common)	Covalent (π bonds common)	Covalent	None
Bonding with metals	Ionic/polar covalent; anions with active metals	Ionic	Ionic	None
Acid-base behavior of common oxide	Strongly acidic (NO_2)	—	Acidic	None
Redox behavior (O.N.)	Every oxidation state from +5 to -3	O_2 (and O_3) very strong oxidizing agents (-2)	Strongest oxidizing agent (-1)	None
	Component of proteins; ammonia for fertilizers, explosives; oxides involved in manufacturing and air pollution (smog, acid rain)	Final oxidizer in residential, industrial, and biological energy production	Manufacture of coatings (Teflon); glass etching (HF); refrigerants involved in ozone depletion (CFCs); dental protection (NaF, SnF_2)	Electrified gas in advertising signs

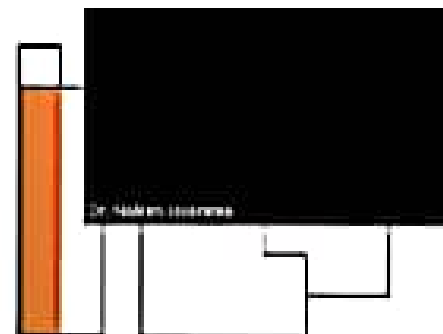


14.3 GROUP 1A(1): THE ALKALI METALS

- The first group of elements in the periodic table form **alkaline (basic) solutions** when their oxides dissolved in water.
- All the elements in the group-lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and rare, radioactive francium (Fr)*are **very reactive** metals.

Why Are the Alkali Metals Soft, Low Melting, and Lightweight?

- Unlike most metals, the **alkali metals are soft**
 - Na has the consistency of cold butter,
 - K can be squeezed like clay.
- The **alkali metals have lower melting and boiling points** than any other group of metals.
 - Except for Li, they all melt below 100°C
 - Cs melts a few degrees above room temperature
- They **have lower densities** than most metals
 - Li floats on lightweight household oil.



GROUP 1A(1)



Lithium floating in oil floating on water.

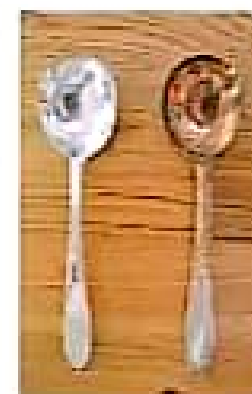
Why Are the Alkali Metals So Reactive?

- The alkali metals are extremely reactive elements.
- They are powerful reducing agents and, thus, always occur in nature as 1+ cations rather than as free metals.
 - The alkali metals reduce the halogens and form ionic solids in reactions that release large quantities of heat.
 - They reduce the hydrogen in water, reacting vigorously (Rb and Cs explosively) to form H_2 and a metal hydroxide solution.
 - They reduce O_2 in the air, and thus tarnish rapidly.

ns^1 configuration is the basis for their physical properties

1. Low heat of atomization (ΔH_{atom}^*).

Consistent with the alkali metals' low melting and boiling points, their weak metallic bonding leads to low values for ΔH_{atom}^* (the energy needed to convert the solid into individual gaseous atoms), which decrease down the group:



Because of this reactivity, Na and K are usually kept under mineral oil in the lab and Rb and Cs are handled with gloves under an inert argon atmosphere.

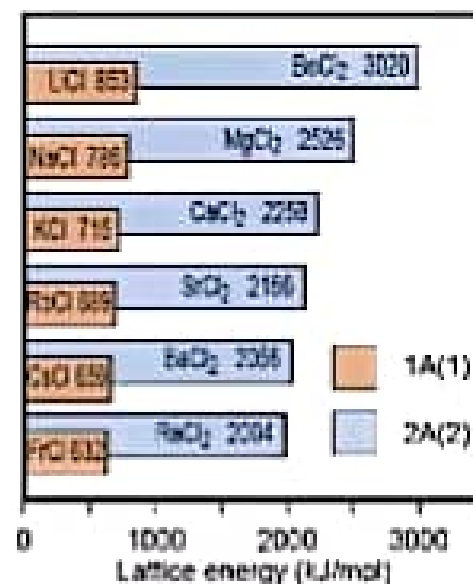
ns^1 configuration is the basis for their physical properties

2. Low IE and small ionic radius.

- Each alkali metal has the largest size and the lowest IE in its period. A great decrease in size occurs when the outer electron is lost.
 - Volume of the Li ion is less than 13% that of the Li atom!
 - Group 1A(1) ions are small spheres with considerable charge density.

3. High lattice energy.

- When Group 1A(1) salts crystallize, large amounts of energy are released because the small cations lie close to the anions.
- For a given anion, the trend in lattice energy is the inverse of the trend in cation size
 - As the cation becomes larger, the lattice energy becomes smaller.




Element	Atomic Number	Ion	Atomic Number
Na	11	Na ⁺	10
K	19	K ⁺	18
Rb	37	Rb ⁺	36
Cs	55	Cs ⁺	54
Fr	87	Fr ⁺	86

- Despite the strong ionic attractions in the solid, nearly all Group 1A (1) salts are water soluble.
- The attraction between the ions and water molecules creates a highly exothermic heat of hydration, and an increase in entropy occurs when ions in the organized crystal become dispersed and hydrated in solution; together, these factors outweigh the high lattice energy.



The magnitude of the hydration energy *decreases* as ionic size increases:



- Interestingly, the smaller ions attract water molecules strongly enough to form larger hydrated ions. This size trend has a major effect on the function of nerves, kidneys, and cell membranes because the sizes of $\text{Na}^+(\text{aq})$ and $\text{K}^+(\text{aq})$, the most common cations in cell fluids, influence their movement in and out of cells.

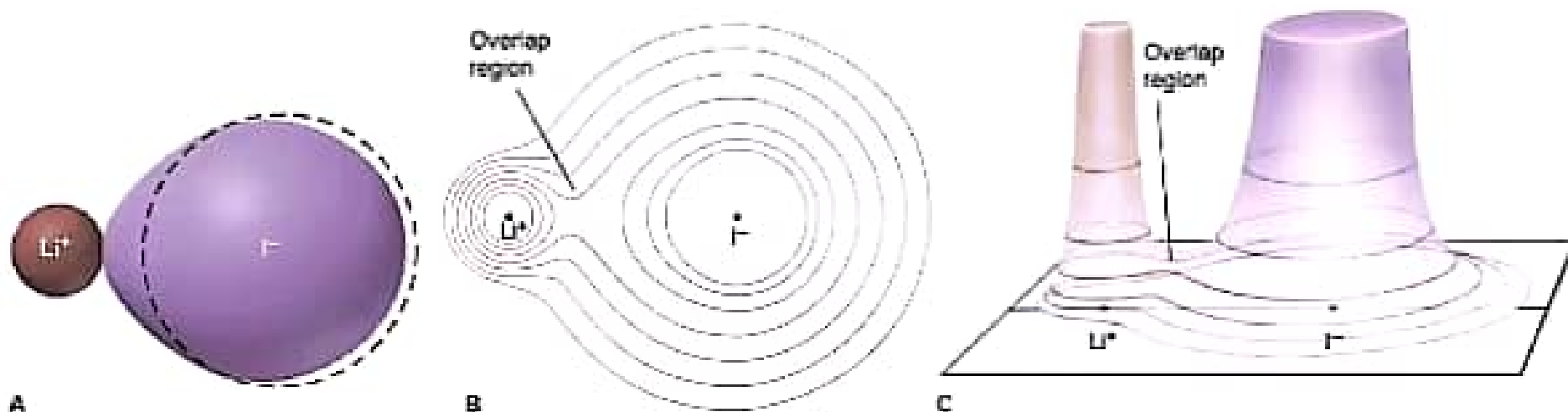


The Anomalous Behavior of Lithium

- It is the only member that forms a simple oxide and nitride, Li_2O and Li_3N , on reaction with O_2 and N_2 in
- Only Li forms molecular compounds with hydrocarbon groups from organic halides:



- Because of its small size, Li has a relatively high charge density.
 - Therefore, it can deform nearby electron clouds to a much greater extent than the other 1A ions can, which increases orbital overlap and gives many lithium salts significant covalent character



- Thus, LiCl, LiBr, and LiI are much more soluble in polar organic solvents, such as ethanol and acetone, than those of Na and K, because the lithium halide dipole interacts with these solvents through dipole-dipole forces.
- The small, highly positive Li makes dissociation of Li salts into ions more difficult in water; thus, the fluoride, hydroxide, and phosphate of Li are much less soluble in water than those of Na and K.



Important Compounds

1. Lithium chloride and lithium bromide, LiCl and LiBr.

Because the Li ion is so small, Li salts have a high affinity for H_2O and yet a positive heat of solution. Thus, they are used in dehumidifiers and air-cooling units.



2. Lithium carbonate, Li_2CO_3 .

Used to make porcelain enamels and toughened glasses and as a drug in the treatment of bipolar disorders.



3. Sodium chloride, NaCl.

Millions of tons used in the industrial production of Na, NaOH, Na_2CO_3 /NaHCO₃, Na_2SO_4 , and HCl or purified for use as table salt.



4. Sodium carbonate and sodium hydrogen carbonate, Na_2CO_3 and NaHCO_3 .

Carbonate used as an industrial base and to make glass. Hydrogen carbonate, which releases CO_2 at low temperatures (50°C to 100°C), used in baking powder and in fire extinguishers.



5. Sodium hydroxide, NaOH .

Most important industrial base; used to make bleach, sodium phosphates, and alcohols.

6. Potassium nitrate, KNO_3 .

Powerful oxidizing agent used in gunpowder and fireworks



Important Reactions

The reducing power of the alkali metals (E) is shown in reactions 1 to 4. Some industrial applications of Group 1A(1) compounds are shown in reactions 5 to 7.

1. The alkali metals reduce H in H_2O from the +1 to the 0 oxidation state:



The reaction becomes more vigorous down the group (see photo).



Potassium reacting with water

2. The alkali metals reduce oxygen, but the product depends on the metal. Li forms the oxide, Li₂O; Na forms the peroxide, Na_2O_2 ; K, Rb, and Cs form the superoxide, EO_2 :



In emergency breathing units, KO_2 reacts with H_2O and CO_2 in exhaled air to release O_2 .

3. The alkali metals reduce hydrogen to form ionic (s) hydrides:



NaH is an industrial base and reducing agent that is used to prepare other reducing agents, such as NaBH_4 .

4. The alkali metals reduce halogens to form ionic halides:



5. Sodium chloride is the most important alkali metal halide. (a) In the Downs process for the production of sodium metal (Section 22.4), reaction 4 is reversed by supplying electricity to molten NaCl :



(b) In the chlor-alkali process (Section 22.5), $\text{NaCl}(aq)$ is electrolyzed to form several key industrial chemicals:



(c) In its reaction with sulfuric acid, NaCl forms two major products:



Sodium sulfate is important in the paper industry; HCl is essential in steel, plastics, textile, and food production.

6. Sodium hydroxide is used in the formation of bleaching solutions:



7. In an ion-exchange process (Chemical Connections, p. 542), water is "softened" by removal of dissolved hard-water cations, which displace Na^+ from a resin:



14.4 GROUP 2A(2): THE ALKALINE EARTH METALS

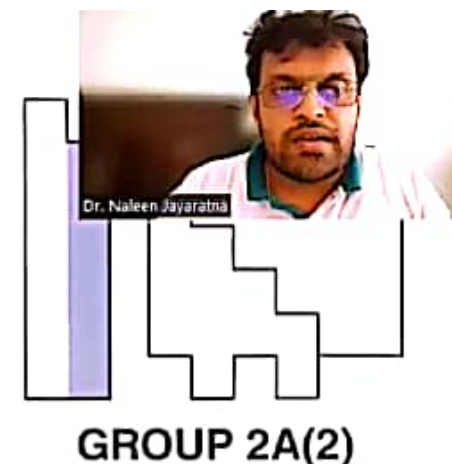
The Group 2A(2) elements are called *alkaline earth metals* because

- Their oxides give basic (alkaline) solutions
- Melt at high temperatures that they remained as solids ("earths").

How Do the Physical Properties of the Alkaline Earth and Alkali Metals Compare?

In general terms, the elements in Groups 1A(1) and 2A(2) behave as close cousins. The differences occur between the groups are due to the change in outer electron configuration: ns^2 vs. ns^1 .

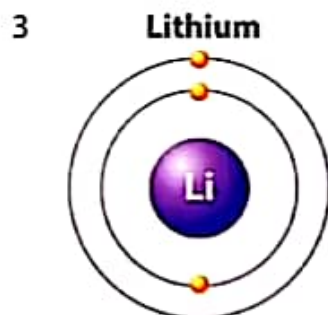
- Two electrons are available from each 2A atom for metallic bonding, and the nucleus contains one additional positive charge.
- These factors make the attraction between delocalized electrons and atom cores greater.
- Consequently, melting and boiling points are much higher for 2A metals than for the corresponding 1A metals
- In fact, the 2A elements melt at around the same temperatures as the 1A elements boil!
- Compared with transition metals, such as iron and chromium, the alkaline earths are soft and lightweight, but they are much harder and more denser than the alkali metals.



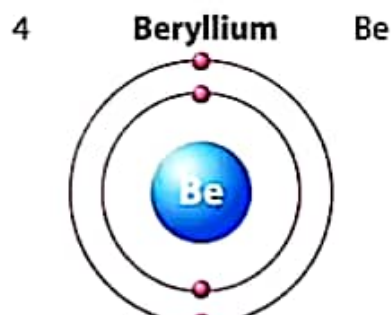


How Do the Chemical Properties of the Alkaline Earth and Alkali Metals Compare?

- The alkaline earth metals display a wider range of chemical behavior than the alkali metals, largely because of the behavior of beryllium.

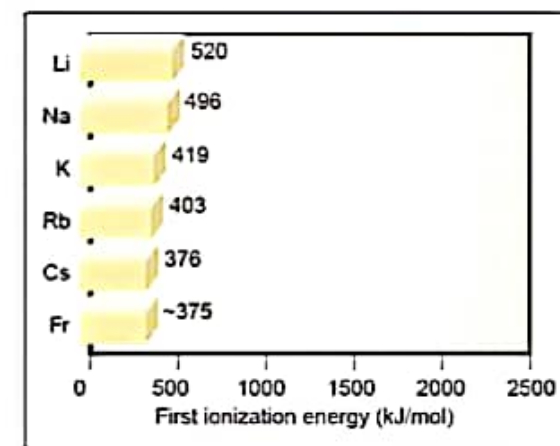
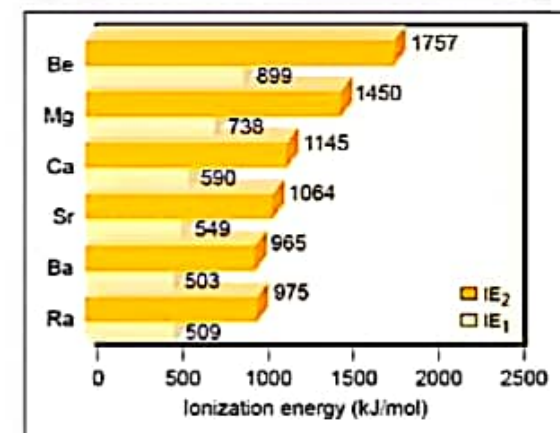


Atomic mass: 6.94
Electron configuration: 2, 1



Atomic mass: 9.0121
Electron configuration: 2, 2

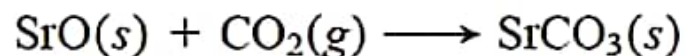
Atomic radius (pm)	Ionic radius (pm)	Atomic radius (pm)	Ionic radius (pm)
Be 112		Li 152	Li ⁺ 76
Mg 160	Mg ²⁺ 72	Na 186	Na ⁺ 102
Ca 197	Ca ²⁺ 100	K 227	K ⁺ 138
Sr 215	Sr ²⁺ 118	Rb 248	Rb ⁺ 152
Ba 222	Ba ²⁺ 135	Cs 265	Cs ⁺ 167
Ra (~220)	Ra ²⁺ 148	Fr (~270)	Fr ⁺ 180



- Because the second valence electron lies in the same sublevel as the first, it is not shielded from the additional nuclear charge very well, and so Z_{eff} is greater. Therefore, Group 2A(2) elements have smaller atomic radii and higher ionization energies than Group 1A(1) elements.

How Do the Chemical Properties of the Alkaline Earth and Alkali Metals Compare?

- Despite the higher IEs, *all the alkaline earths (except Be) form ionic compounds as 2 cations.*
- Beryllium behaves differently because so much energy is needed to remove two electrons from this tiny atom that it never forms discrete Be^{2+} ions, and its bonds are polar covalent.
- Like the alkali metals, the alkaline earth metals are strong reducing agents. Each element reduces O_2 in air to form the oxide (Ba also forms the peroxide, BaO_2).
- Except for Be and Mg, which form adherent oxide coatings, the alkaline earths reduce H_2O at room temperature to form H_2 .
- And, except for Be, they reduce the halogens, N_2 , and H_2 to form ionic compounds.
- The Group 2A oxides are strongly basic (except for amphoteric BeO) and react with acidic oxides to form salts, such as sulfites and carbonates; for example,

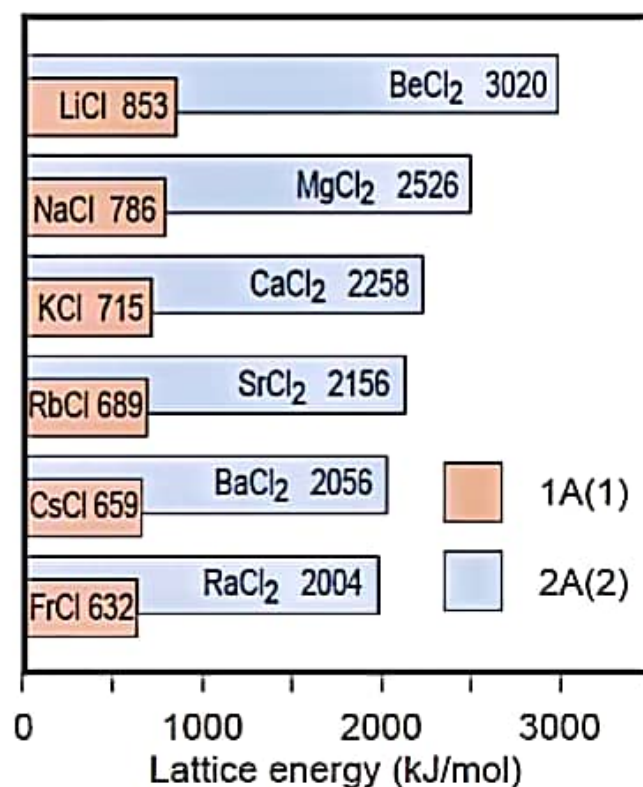


- Natural carbonates, such as limestone and marble, are major structural materials and the commercial sources for most 2A compounds.



160		72
Ca		Ca^{2+}
197		100
Sr		Sr^{2+}
215		118
Ba		Ba^{2+}
222		135
Ra		Ra^{2+}
(~220)		148

- Group 2A salts have much higher lattice energies than Group 1A salts because the 2A cations are smaller and more highly charged.
- One of the main differences between the two groups is the lower solubility of 2A salts in water. Their ions are smaller and more highly charged than 1A ions, resulting in much higher charge densities. Therefore, stronger the electrostatic attraction among each other, harder the water molecule to pull them apart.



The Anomalous Behavior of Beryllium (Differ from the rest of the members of the group)



Main reasons are;

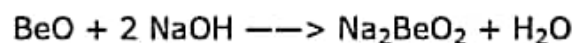
- Exceptionally small atomic and ionic size
- High ionization enthalpy
- Absence of d orbitals in its valence shell

Some important properties in which beryllium differ from rest of the members

- Harder than other members
- Higher boiling and melting points
- Does not react with water even at high temperature, while others do



- Forms covalent compounds whereas other members form ionic compounds
- Beryllium oxides and hydroxides are amphoteric while other oxides and hydroxides of other alkaline earth metals are basic.

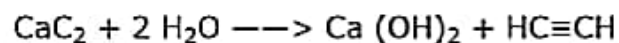
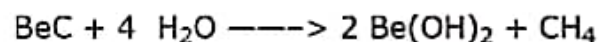


The Anomalous Behavior of Beryllium (Differ from the rest of the members of the group)



Some important properties in which beryllium differ from rest of the members

- carbides of Be are covalent whereas carbides of other members are ionic
- Be carbide react with water to produce methane gas, carbides of other members form acetylene gas

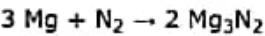
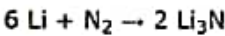


- Beryllium does not exhibit coordination number of more than four since its valence shell has only 4 orbitals, one s and p, other elements in the group can show coordination number 6 using s, p and d orbitals

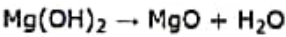
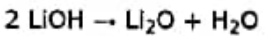
Diagonal Relationships: Lithium and Magnesium

	1A (1)	2A (2)	3A (13)	4A (14)
2	Li	Be	B	
3		Mg	Al	Si

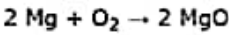
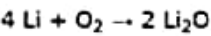
- (1) Both have almost **similar electronegativities**.
- (2) Both Li and Mg are **quite hard**. They are harder and lighter than other elements in their respective groups.
- (3) Both LiOH and Mg(OH)₂ are **weak bases**.
- (4) Both **form ionic nitrides** when heated in atmosphere of Nitrogen.



- (5) The **hydroxides** of both lithium and magnesium **decompose upon heating**.

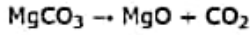
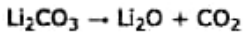


- (6) Both lithium and magnesium **combine with oxygen** to form oxides. Members of their respective groups form peroxide and superoxide.



The Li₂O and MgO thus formed do not combine with excess O₂ to form peroxide and superoxide.

- (7) The **carbonates** of these metals **decompose on heating** to the corresponding **oxides** with the evolution of **carbon dioxide**.



- (8) Both Lithium and magnesium **do not form solid bicarbonates**.

- (9) Both Lithium and magnesium **nitrate decompose on heating** producing nitrogen dioxide.



- (10) The hydroxides, carbonate, oxalates, phosphates and fluorides of both Lithium and magnesium are sparingly soluble in water.

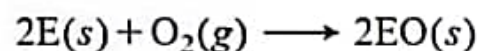


Dr. Nalveen Jayaratna

Important Reactions

The elements (E) act as reducing agents in reactions 1 to 5; note the similarity to reactions of Group 1A(1). Reaction 6 shows the general basicity of the 2A(2) oxides; reaction 7 shows the general instability of their carbonates at high temperature.

1. The metals reduce O_2 to form the oxides:

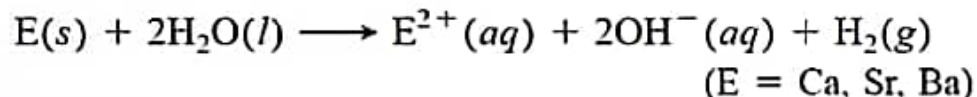


Ba also forms the peroxide, BaO_2 .



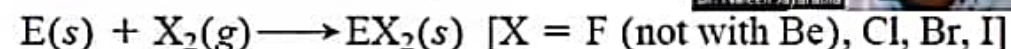
Magnesium ribbon burning

2. The larger metals reduce water to form hydrogen gas:

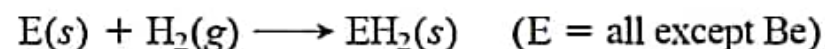


Be and Mg form an adherent oxide coating that allows only slight reaction.

3. The metals reduce halogens to form ionic halides:



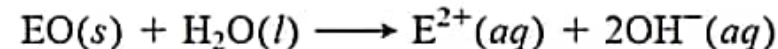
4. Most of the elements reduce hydrogen to form ionic hydrides:



5. Most of the elements reduce nitrogen to form ionic nitrides:

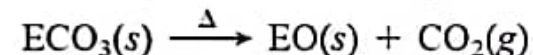


6. Except for amphoteric BeO , the element oxides are basic:



$Ca(OH)_2$ is a component of cement and mortar.

7. All carbonates undergo thermal decomposition to the oxide:



This reaction is used to produce CaO (lime) in huge amounts from naturally occurring limestone (*margin note, p. 574*).



Dr. Naleen Jayaratna

Important Compounds



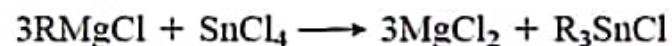
1. Beryl, $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$. Beryl, the industrial source of Be metal, also occurs as a gemstone with a variety of colors (*see photo*). It is chemically identical to emerald, except for the trace of Cr^{3+} that gives emerald its green color.

2. Magnesium oxide, MgO . Because of its high melting point (2852°C), MgO is used as a refractory material for furnace brick (*see photo*) and wire insulation.

Industrial kiln



3. Alkylmagnesium halides, RMgX (R = hydrocarbon, X = halogen). These compounds are used to synthesize many organic compounds. Organotin agricultural fungicides are made by treating RMgX with SnCl_4 :



4. Calcium carbonate, CaCO_3 . Occurs as enormous natural deposits of limestone, marble, chalk (*see photo*), and coral. Used as a building material, to make lime, and, in high purity, as a tooth-paste abrasive and an antacid.

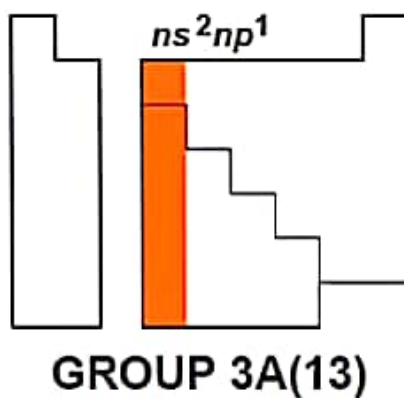
White cliffs,
Dover, England



14.5 GROUP 3A(13): THE BORON FAMILY

Group 3A(13) is the first of the p block

KEY
Atomic No.
Symbol
Atomic mass
Valence e ⁻ configuration
Common oxidation states



5 B 10.81 $2s^2 2p^1$ +3	
13 Al 26.98 $3s^2 3p^1$ +3	
31 Ga 69.72 $4s^2 4p^1$ +3, +1	
49 In 114.8 $5s^2 5p^1$ +3, +1	
81 Tl 204.4 $6s^2 6p^1$ +1	
113 (284) $7s^2 7p^1$	Observed in experiments at Dubna, Russia, in 2003

Meleesha

liashakaumini@gmail.com

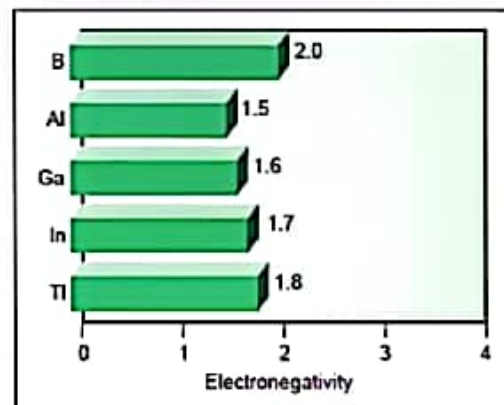
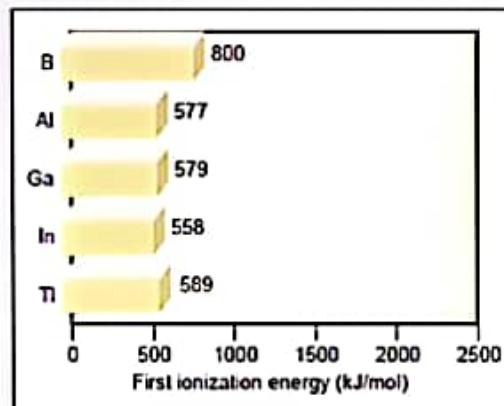
Atomic Properties

Atomic radius (pm)		Ionic radius (pm)
B 85		
Al 143		Al ³⁺ 54
Ga 135		Ga ³⁺ 62
In 167		In ³⁺ 80
Tl 170		Tl ⁺ 150

Group electron configuration is ns^2np^1 . All except Tl (and presumably element 113) commonly display the +3 oxidation state. The +1 state becomes more common down the group.

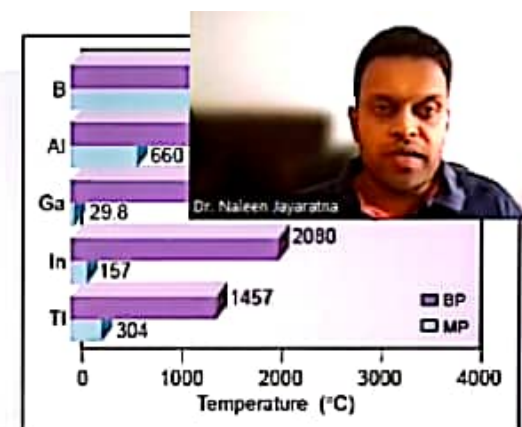
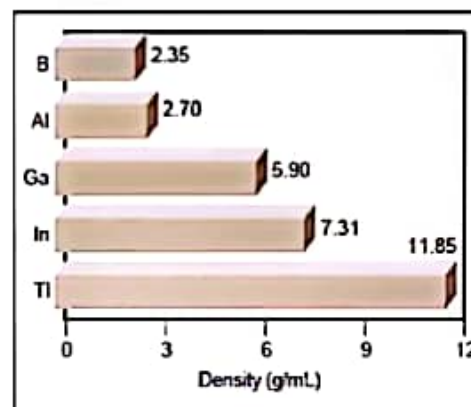
Atomic size is smaller and EN is higher than for 2A(2) elements; IE is lower, however, because it is easier to remove an electron from the higher energy p sublevel.

Atomic size, IE, and EN do not change as expected down the group because there are intervening transition and inner transition elements.



Physical Properties

Bonding changes from network covalent in B to metallic in the rest of the group. Thus, B has a much higher melting point than the others, but there is no overall trend. Boiling points decrease down the group.



Densities increase down the group.

The redox behavior of the elements in the group



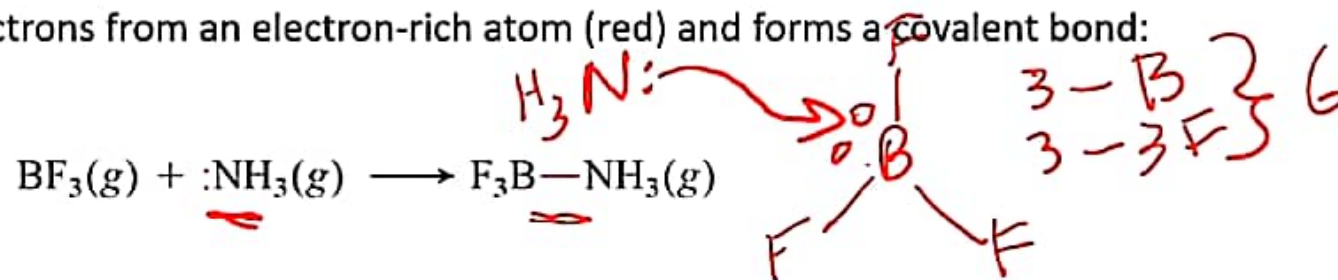
1. multiple oxidation states present.
2. lower state becomes more prominent going down the group.
3. oxides with the element in a lower oxidation state are more basic than oxides with the element in a higher oxidation state.

Highlights of Boron Chemistry

$s^2 p^1$ 3

Accepting a Bonding Pair from an Electron-Rich Atom

In gaseous boron trihalides (BX_3), the B atom is electron deficient, with only six electrons around it. To attain an octet, the B atom accepts a lone pair of electrons from an electron-rich atom (red) and forms a covalent bond:



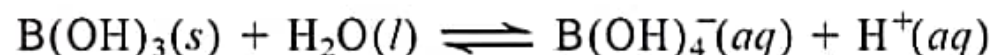
Such reactions, in which one reactant accepts an electron pair from another to form a covalent bond are known as Lewis acid-base reactions.

Highlights of Boron Chemistry



Dr. Nalini Jayaraj

- Similarly, B has only six electrons in boric acid, B(OH)_3 . In water, the acid itself does not release a proton. Rather, it accepts an electron pair from the O in H_2O , forming a fourth bond and releasing an H^+ ion:

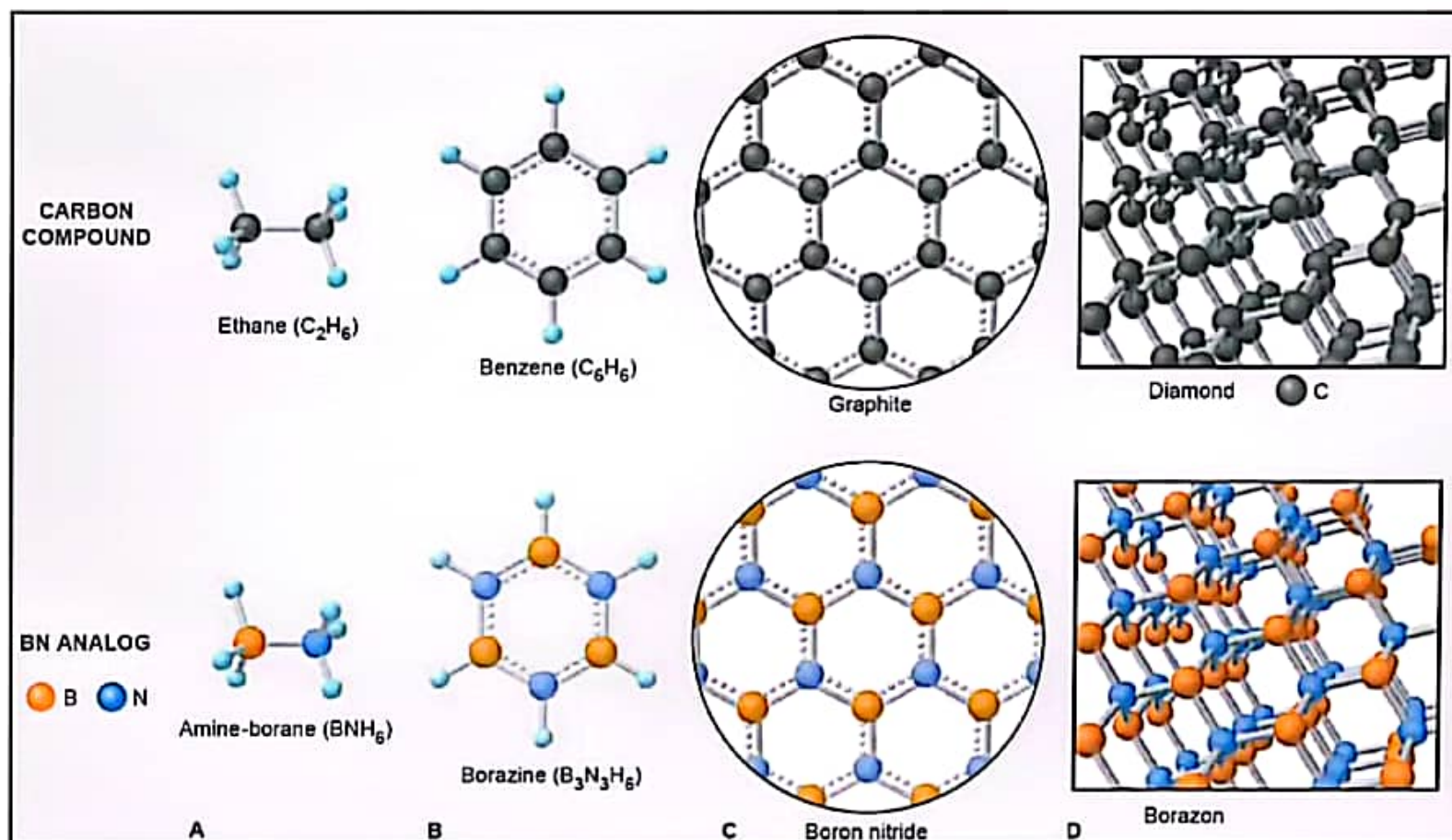


- Boron also attains an octet in several boron-nitrogen compounds whose crystal or molecular structures are amazingly similar to those of elemental carbon and some of its organic compounds. These similarities are due to the fact that the size, IE, and EN of C are between those of B and N.
- Moreover, C has four valence electrons $\cdot\dot{\text{C}}-\dot{\text{C}}\cdot$ and $\cdot\dot{\text{B}}-\dot{\text{N}}\cdot$. N has five, so the following grouping have the same number of valence electrons.

Therefore, $\text{H}_3\text{C}-\text{CH}_3$ (ethane) and $\text{H}_3\text{B}-\text{NH}_3$ (amine-borane) are isoelectronic.

Highlights of Boron Chemistry

- Boron nitride has a structure consisting of hexagons fused into sheets, very similar to that of graphite. In graphite, its electrons are highly delocalized through resonance.

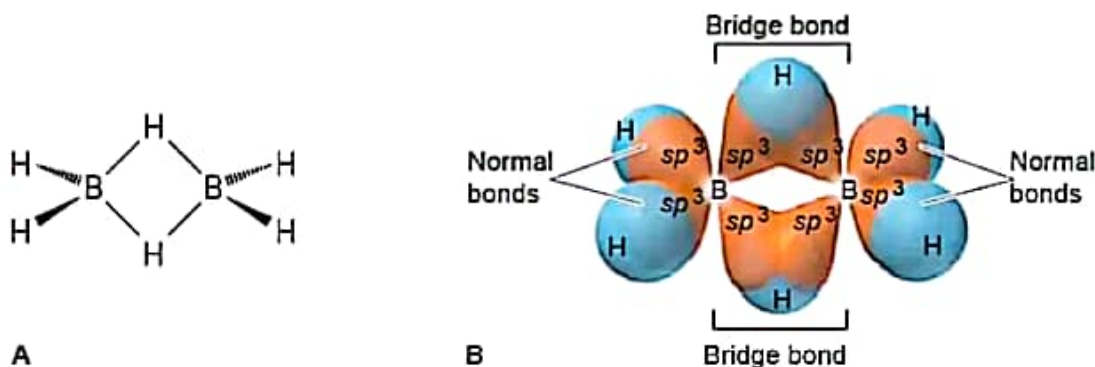


Highlights of Boron Chemistry



Forming Bridge Bonds with Electron-Poor Atoms

- In elemental boron and its many hydrides (boranes), there is no electron-rich atom to supply boron with electrons. In these substances, boron attains an octet through some unusual bonding.
- In diborane (B_2H_6) and many larger boranes, for example, two types of B-H bonds exist.
 - The first type is a typical electron-pair bond.



- A sp^3 orbital of B overlapping a $1s$ orbital of H in each of the four terminal B-H bonds, using two of the three electrons in the valence level of each B atom.

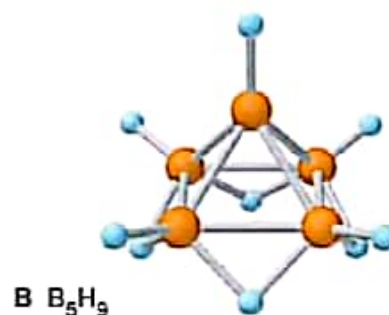
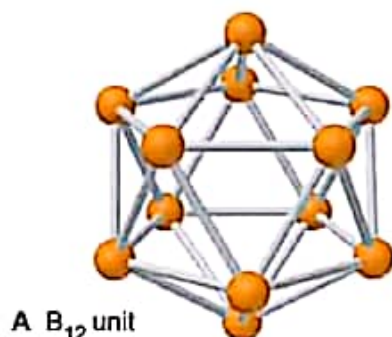
Highlights of Boron Chemistry



- Other type of bond is a hydride bridge bond (or three-center, two electron bond)

Each B-H-B grouping is held together by only two electrons. Two sp^3 orbitals, one from each B, overlap an H 1s orbital between them.

- Two electrons move through this extended bonding orbital—one from one of the B atoms and the other from the H atom—and join the two B atoms via the H atom bridge.
- Notice that each B atom is surrounded by eight electrons: four from the two normal B-H bonds and four from the two B-H-B bridge bonds.
- In many boranes and in elemental boron, one B atom bridges two others in a three-center, two-electron B-B-B bond.



Diagonal Relationships: Beryllium and Aluminum



- Beryllium in Group 2A(2) and aluminum in Group 3A(13) are another pair of diagonally related elements. Both form oxoanions in strong base:

beryllate, $\text{Be}(\text{OH})_4^{2-}$, and aluminate, $\text{Al}(\text{OH})_4^{2-}$

- Both have bridge bonds in their hydrides and chlorides.
- Both form oxide coatings impervious to reaction with water, and both oxides are amphoteric, extremely hard, and high melting.
- Although the atomic and ionic sizes of these elements differ, the small, highly charged Be^{2+} and Al^{3+} ions polarize nearby electron clouds strongly. Therefore, some Al compounds and all Be compounds have significant covalent character.



Dr. Nalren Jayaratna

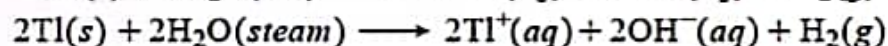
Some Reactions and Compounds



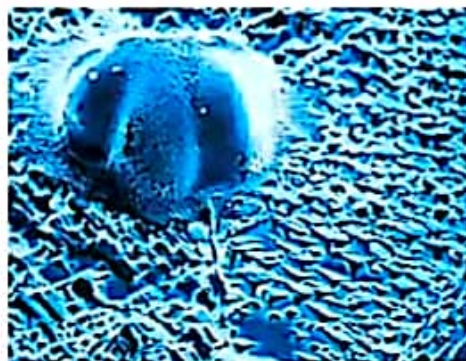
Important Reactions

In reactions 1 to 3, the elements (E) usually require higher temperatures to react than those of Groups 1A(1) and 2A(2); note the lower oxidation state of Tl. Reactions 4 to 6 show some compounds in key industrial processes.

1. The elements react sluggishly, if at all, with water:

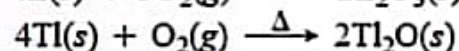
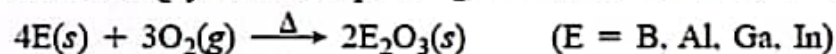


Al becomes covered with a layer of Al_2O_3 that prevents further reaction (see photo).



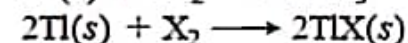
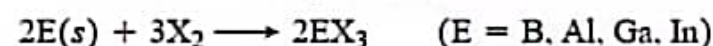
Electron micrograph of oxide coating on Al surface

2. When strongly heated in pure O_2 , all members form oxides:



Oxide acidity decreases down the group: B_2O_3 (weakly acidic) $>$ $\text{Al}_2\text{O}_3 > \text{Ga}_2\text{O}_3 > \text{In}_2\text{O}_3 > \text{Tl}_2\text{O}$ (strongly basic), and the +1 oxide is more basic than the +3 oxide.

3. All members reduce halogens (X_2):



The BX_3 compounds are volatile and covalent.

Trihalides of Al, Ga, and In are (mostly) ionic solids but occur as covalent dimers in the gas phase; in this way, the 3A atom attains a filled outer level.

4. Acid treatment of Al_2O_3 is important in water purification:



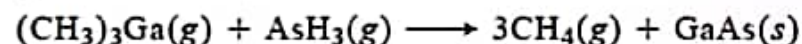
In water, $\text{Al}_2(\text{SO}_4)_3$ and other agents form a colloid that aids in removing suspended particles (see *Chemical Connections*, p. 541).

5. The overall reaction in the production of aluminum metal is a redox process:



This process is carried out electrochemically in the presence of cryolite (Na_3AlF_6), which lowers the melting point of the reactant mixture and takes part in the change (Section 22.4).

6. A displacement reaction produces gallium arsenide, GaAs (margin note, p. 578):



Important Compounds

1. Boron oxide, B_2O_3 . Used in the production of borosilicate glass (*see margin note, p. 582*).
2. Borax, $Na_2[B_4O_5(OH)_4] \cdot 8H_2O$. Major mineral source of boron compounds and B_2O_3 . Used as a fireproof insulation material and as a washing powder (20-Mule Team Borax).
3. Boric acid, H_3BO_3 [or $B(OH)_3$]. Used as external disinfectant, eyewash, and insecticide.
4. Diborane, B_2H_6 . A powerful reducing agent for possible use as a rocket fuel. Used to synthesize higher boranes, compounds that led to new theories of chemical bonding.



5. Aluminum sulfate (cake alum), $Al_2(SO_4)_3$. Used in purifying water, tanning leather, and sizing paper; as a fixative for dyeing cloth (*see photo*); and as an antiperspirant.

6. Aluminum oxide, Al_2O_3 . Major compound in (bauxite) of Al metal. Used as abrasive in sandpaper, sanding and cutting tools, and toothpaste. Large crystals with metal ion impurities often of gemstone quality (*see photo*). Inert support for chromatography. In fibrous forms, woven into heat-resistant fabrics; also used to strengthen ceramics and metals.



7. $Tl_2Ba_2Ca_2Cu_3O_{10}$. Becomes a high-temperature superconductor at 125 K, which is readily attained with liquid N_2 (*see photo*).



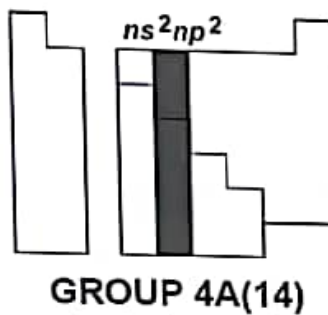
Magnet levitated
by superconductor



Dr. Naleen Jayaratna

GROUP 4A(14): THE CARBON FAMILY

KEY
Atomic No.
Symbol
Atomic mass
Valence e ⁻ configuration
Common oxidation states



6 C 12.01 $2s^2 2p^2$ (-4, +4, -2)	
14 Si 28.09 $3s^2 3p^2$ (-4, +4)	
32 Ge 72.61 $4s^2 4p^2$ (+4, -2)	
50 Sn 118.7 $5s^2 5p^2$ (+4, -2)	
82 Pb 207.2 $6s^2 6p^2$ (+4, +2)	
114 (285) $7s^2 7p^2$	Observed in experiments at Dubna, Russia, in 1998



Table 14.2 Bond Type and the Melting Process in Groups 3A(13) to 5A(15)

Period	Group 3A(13)				Group 4A(14)				Group 5A(15)				Key:
	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	
2	B		2180	23.6	C		4100	Very high	N		-210	0.7	 Metal  Covalent network  Covalent molecule  Metal  Metalloid  Nonmetal
3	Al		660	10.5	Si		1420	50.6	P		44.1	2.5	
4	Ga		30	5.6	Ge		945	36.8	As		816	27.7	
5	In		157	3.3	Sn		232	7.1	Sb		631	20.0	
6	Tl		304	4.3	Pb		327	4.8	Bi		271	10.5	



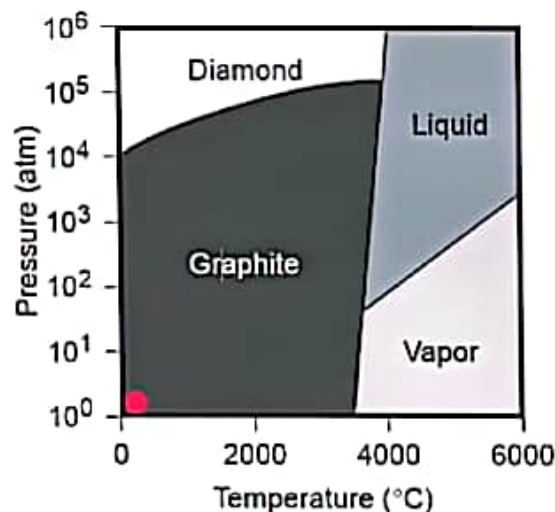
- Within Group 4A, the large decrease in melting point between the network covalent solids C and Si is due to longer, weaker bonds in the Si structure and the large decrease between Ge and Sn is due to the change from covalent network to metallic bonding.
- Considering horizontal trends, the large increases in melting point and ΔH_{fus} across a period between Al and Si and between Ga and Ge reflect the change from metallic to covalent network bonding.
- Note the unexpected rises in the values for these properties from metallic Al, Ga, and Sn to the network covalent metalloids Si, Ge, and Sb, and note the abrupt drops from the covalent networks of C and Si to the individual molecules of N and P in Group 5A.



Allotropism: Different Forms of an Element

Allotropes: different crystalline or molecular forms of a substance.

One allotrope is usually more stable than another at a particular pressure and temperature.



- Graphite is a black electrical conductor that is soft and “greasy,” whereas diamond is a colorless electrical insulator that is extremely hard.
- Graphite is the standard state of carbon, the more stable form at ordinary temperature and pressure.
- Fortunately for jewelry owners, diamond changes to graphite at a negligible rate under normal conditions.

Figure 14.12 Phase diagram of carbon. Graphite is the more stable form of carbon at ordinary conditions (*small red circle at extreme lower left*). Diamond is more stable at very high pressure.

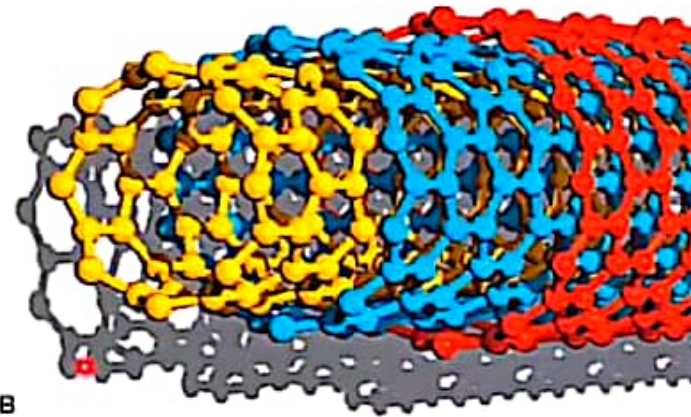
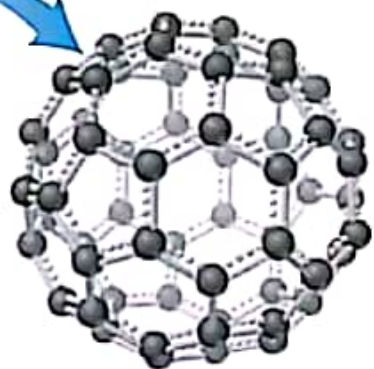


Figure 14.
nanotubes.

fullerene (C₆₀) model. The "buckyball," is a soccer ball-shaped molecule of 60 carbon atoms. **B**, Nanotubes are single or, as shown in this colored model, concentric graphite-like tubes (shown here without the usual fullerene ends; see the Gallery, p. 398).



Tin (Sn) has two allotropes

- White -tin is stable at room temperature and above, whereas gray -tin is the more stable form below 13°C.
- When white tin is kept for long periods at a low temperature, some converts to microcrystals of gray tin.
- The random formation and growth of these regions of gray tin, which has a different crystal structure, weaken the metal and make it crumble.

Group 7A(17): The Halogens

KEY

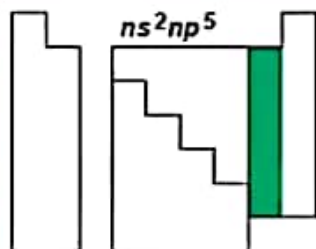
Atomic No.

Symbol

Atomic mass

Valence e⁻
configuration

Common
oxidation
states



GROUP 7A(17)

Atomic Properties

Atomic
radius
(pm)

F

72



Cl

100



Br

114



I

133



At

(140)



F⁻

133

Cl⁻

181

Br⁻

196

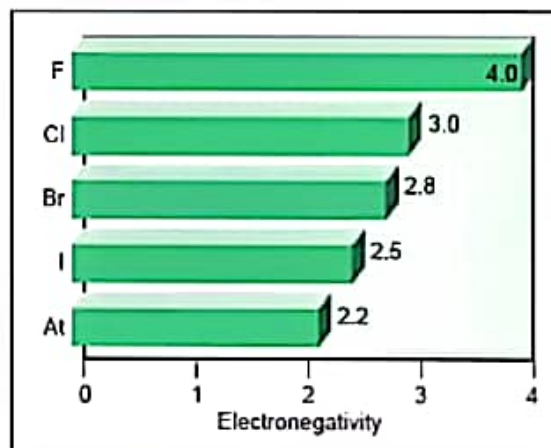
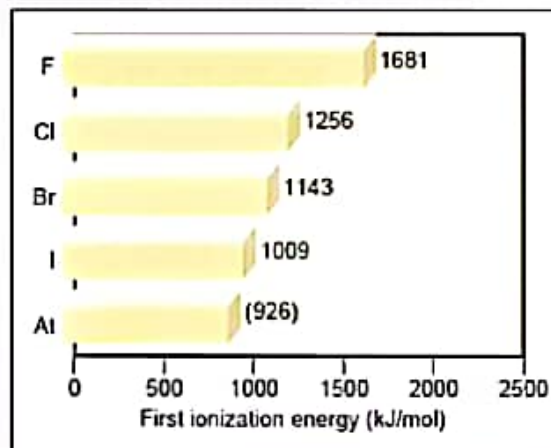
I⁻

220

no data

Group electron configuration is ns^2np^5 ; elements lack one electron to complete their outer level. The -1 oxidation state is the most common for all members. Except for F, the halogens exhibit all odd-numbered states (+7 through -1).

Down the group, atomic and ionic size increase steadily, as IE and EN decrease.



17
Cl

35.45
 $3s^23p^5$
(-1, +7,
+5, +3,
+1)



35
Br

79.90
 $4s^24p^5$
(-1, +7,
+5, +3,
+1)



53
I

126.9
 $5s^25p^5$
(-1, +7,
+5, +3,
+1)



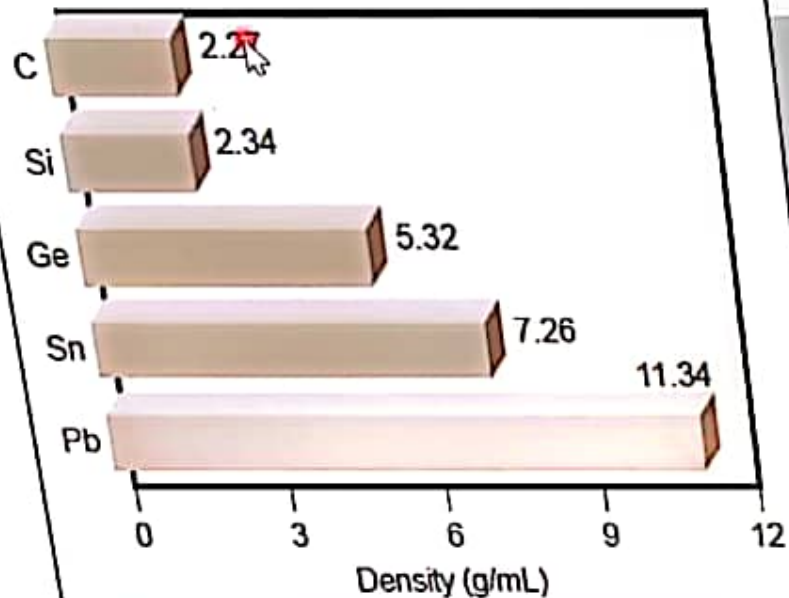
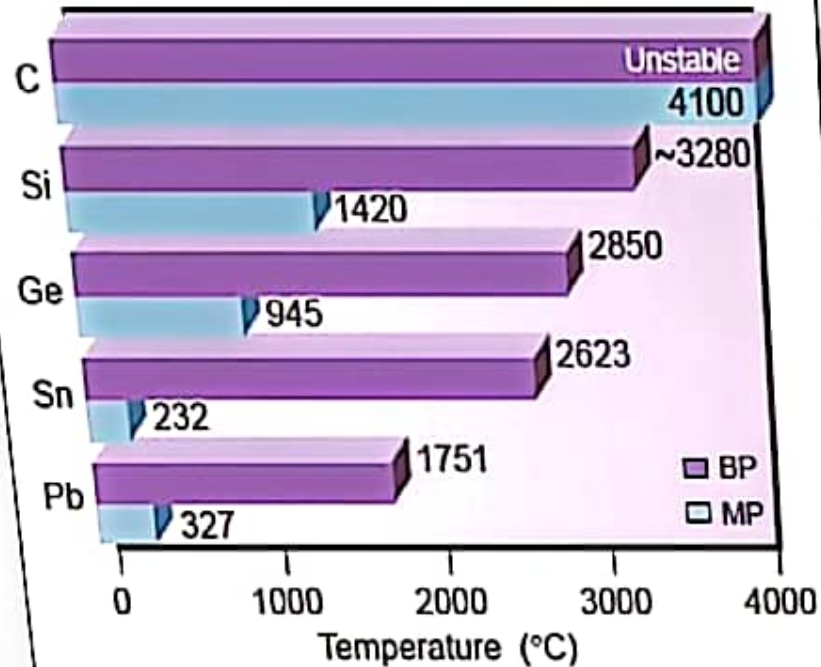
85
At

(210)
 $6s^26p^5$
(-1)

Extremely rare,
no sample available

Physical Properties

Trends in properties, such as decreasing hardness and melting point, are due to changes in types of bonding within the solid: network covalent in C, Si, and Ge; metallic in Sn and Pb (*see text*).



Down the group, density increases because of several factors, including differences in crystal packing.

SOME REACTIONS AND COMPOUNDS

Important Reactions

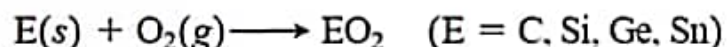
Reactions 1 and 2 pertain to all the elements (E); reactions 3 to 7 concern industrial uses for compounds of C and Si.

1. The elements are oxidized by halogens:

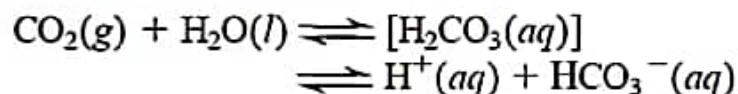


The +2 halides are more stable for tin and lead, SnX_2 and PbX_2 .

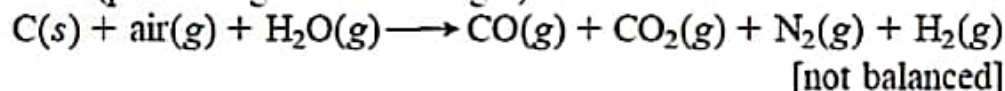
2. The elements are oxidized by O_2 :



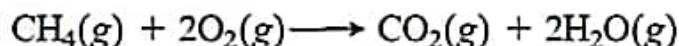
Pb forms the +2 oxide, PbO . Oxides become more basic down the group. The reaction of CO_2 and H_2O provides the weak acidity of natural unpolluted waters:



3. Air and steam passed over hot coke produce gaseous fuel mixtures (producer gas and water gas):

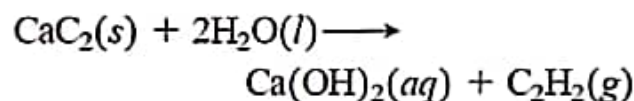


4. Hydrocarbons react with O_2 to form CO_2 and H_2O . The reaction for methane is adapted to yield heat or electricity:



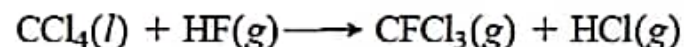
Oxyacetylene torch

5. Certain metal carbides react with water to produce acetylene:



The gas is used to make other organic compounds and as a fuel in welding (see photo).

6. Freons (chlorofluorocarbons) are formed by fluorinating carbon tetrachloride:



Production of trichlorofluoromethane (Freon-11), the most widely used refrigerant in the world, is being eliminated because of adverse effects on the environment (*margin note, p. 589*).

7. Silica is reduced to form elemental silicon:



This crude silicon is made ultrapure through zone refining (*Gallery, p. 533*) for the manufacture of computer chips (see photo).

Computer chip



Important Compounds



1. **Carbon monoxide, CO .** used as a gaseous fuel, as a precursor for organic compounds, as a reactant in the purification of nickel. Formed in internal combustion engines and released as a toxic air pollutant.
2. **Carbon dioxide, CO_2 .** Atmospheric component used by photosynthetic plants to make carbohydrates and O_2 . The final oxidation product of all carbon-based fuels; its increase in the atmosphere is changing Earth's climate. Used industrially as a refrigerant gas, a blanketing gas in fire extinguishers, and an effervescent gas in beverages. Combined with NH_3 to form urea for fertilizers and plastics manufacture.
3. **Methane, CH_4 .** Used as a fuel and in the production of many organic compounds. Major component of natural gas. Formed by anaerobic decomposition of plants (swamp gas) and by microbes in termites and certain mammals. Greenhouse gas that contributes to global warming.

Important Compounds



4. **Silicon dioxide, SiO_2 .** Occurs in many amorphous (glassy) and crystalline forms, with quartz being the most common. Used to make glass and as an inert chromatography support material.
5. **Silicon carbide, SiC .** Known as carborundum, a major industrial abrasive and a highly refractory ceramic for tough, high temperature uses. Can be doped to form a high-temperature semiconductor.
6. **Organotin compounds, R_4Sn .** Used to stabilize PVC [poly(vinyl chloride)] plastics and to cure silicone rubbers.
Agricultural biocide for insects, fungi, and weeds.
7. **Tetraethyl lead, $(\text{C}_2\text{H}_5)_4\text{Pb}$.** Once used as a gasoline additive to improve fuel efficiency, but now banned because of its inactivation of auto catalytic converters. Major source of lead as a toxic air pollutant.

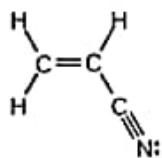




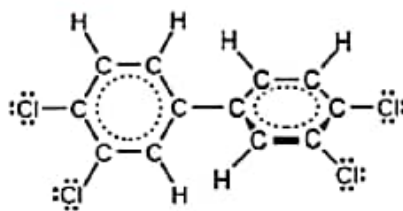
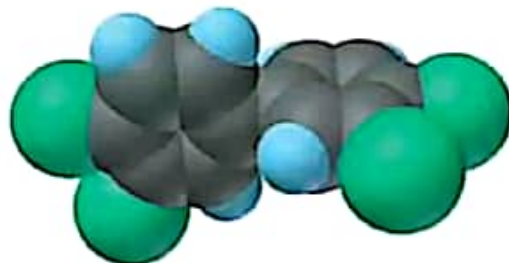
Dr. Nalini Jayaraj

Highlights of Carbon Chemistry

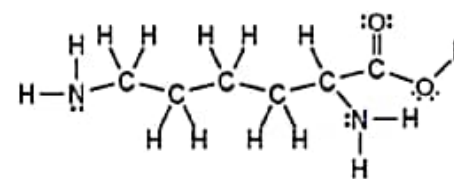
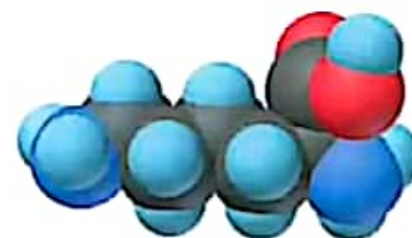
- Carbon forms bonds with the smaller Group 1A(1) and 2A(2) metals, many transition metals, the halogens B, Si, N, O, P, and S.
- It also exhibits every oxidation state possible for its group, from +4 in CO_2 and halides like CCl_4 through -4 in CH_4 .
- Two features stand out in the chemistry of carbon: ability to bond to itself to form chains and to form multiple bonds.
- As a result of its small size and its capacity for four bonds, carbon can form chains, branches, and rings that lead to myriad structures. Add a lot of H, some O and N, a bit of S, P, halogens, and a few metals, and you have the whole organic world!



Acrylonitrile



PCB

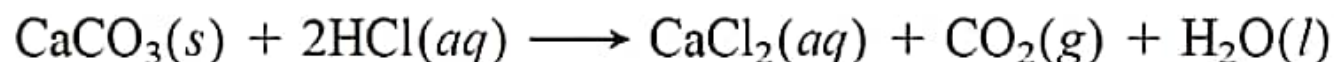


Lysine



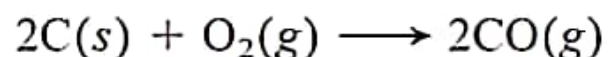
Highlights of Carbon Chemistry

- Multiple bonds are common in carbon structures because the C-C bond is short enough for side-to-side overlap of half-filled 2p orbitals to form π bonds.
- Because the other 4A members are larger, E-E bonds become longer and weaker down the group; thus, in terms of bond strength, $\text{C-C} > \text{Si-Si} > \text{Ge-Ge}$. The empty d orbitals of these larger atoms make the chains much more susceptible to chemical attack.
- Thus, some compounds with long Si chains are known but they are very reactive, and the longest Ge chain has only eight atoms.
- Moreover, longer bonds do not typically allow sufficient overlap of p orbitals for π bonding. Only very recently have compounds with π bonds between Si atoms or between Ge atoms been prepared, and they are also extremely reactive.
- In contrast to its organic compounds, carbon's inorganic compounds are simple. Metal carbonates are the main mineral form. Marble, limestone, chalk, coral, and several other types are found in enormous deposits throughout the world.
- Many of these compounds are leftovers of fossilized marine organisms. Carbonates are used in several common antacids because they react with the HCl in stomach acid

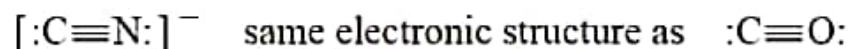




- Unlike the other 4A members, which form only solid network-covalent or ionic oxides, carbon forms two oxides, CO_2 and CO .
- Carbon dioxide is essential to all life: it is the primary source of carbon in plants and animals through photosynthesis.
- Its aqueous solution is the cause of acidity in natural waters.
- However, its atmospheric buildup from deforestation and excessive use of fossil fuels is severely affecting the global climate.
- Carbon monoxide forms when carbon or its compounds burn in an inadequate supply of O_2



- Carbon monoxide is a key component of syngas fuels and is widely used in the production of methanol, formaldehyde, and other major industrial compounds.
- CO binds strongly to many transition metals. When inhaled in cigarette smoke or polluted air, it enters the blood and binds strongly to the Fe(II) in hemoglobin, preventing the normal binding of O_2 , and to other iron-containing proteins.
- The cyanide ion (CN^-) is isoelectronic with CO :





Highlights of Silicon Chemistry

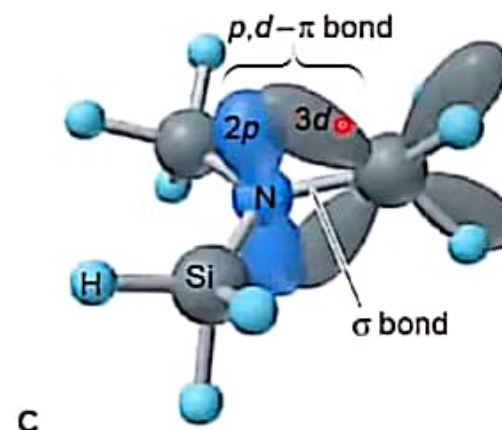
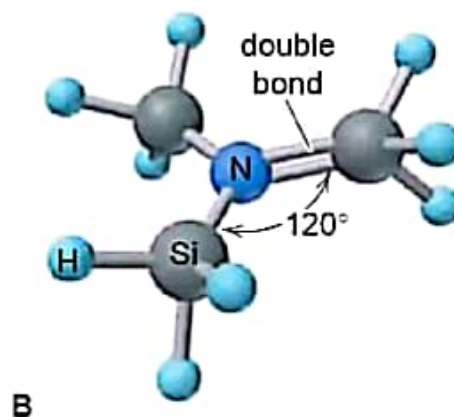
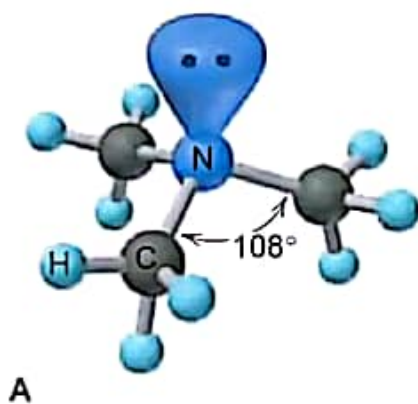
- Silicon halides are more reactive than carbon halides because Si has empty d orbitals that are available for d -orbital formation.
- One explanation for this unusual strength is that the Si-X bond has some double-bond character because of the presence of a σ bond and a different type of π bond, called a $p_{\pi} - d_{\pi}$ bond. This bond arises from side-to-side overlap of an Si d orbital and a halogen p orbital.
- A similar type of $p, d - \pi$ bonding leads to the trigonal planar molecular shape of trisilylamine, in contrast to the trigonal pyramidal shape of trimethylamine.





Highlights of Silicon Chemistry

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- Just as carbon forms unending C-C chains, the Si-O grouping repeats itself endlessly in a wide variety of important minerals on the planet, and in **silicones**, synthetic polymers that have many applications:

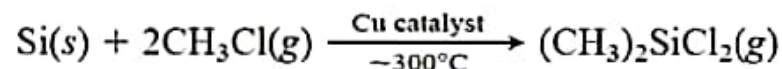
1. Silicate minerals.

- Oxygen, the most abundant element on Earth, and silicon, the next most abundant, compose these minerals and thus account for four of every five atoms on the surface of the planet!

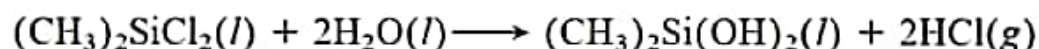


2. Silicone polymers

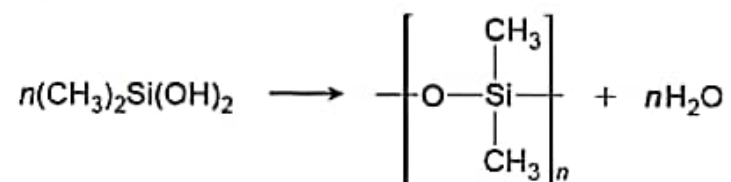
- A key starting material is formed by the reaction of silicon with methyl chloride:



Although the corresponding carbon compound $[(\text{CH}_3)_2\text{CCl}_2]$ is inert in water, the empty d orbitals of Si make $(\text{CH}_3)_2\text{SiCl}_2$ reactive:



The product reacts with itself to form water and a silicone chain molecule called a *poly(dimethyl siloxane)*:



- Silicones have properties of both plastics and minerals. The organic groups give them the flexibility and weak intermolecular forces between chains that are characteristic of a plastic, while the O-Si-O backbone confers the thermal stability and nonflammability of a mineral.



Diagonal Relationships: Boron and Silicon

metalloids boron and silicon. 



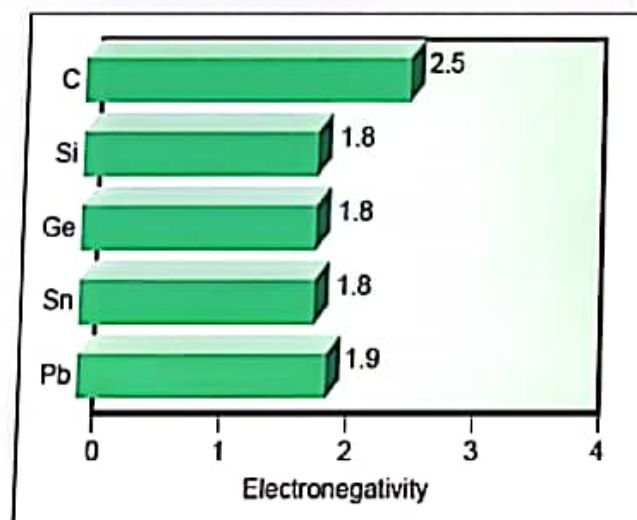
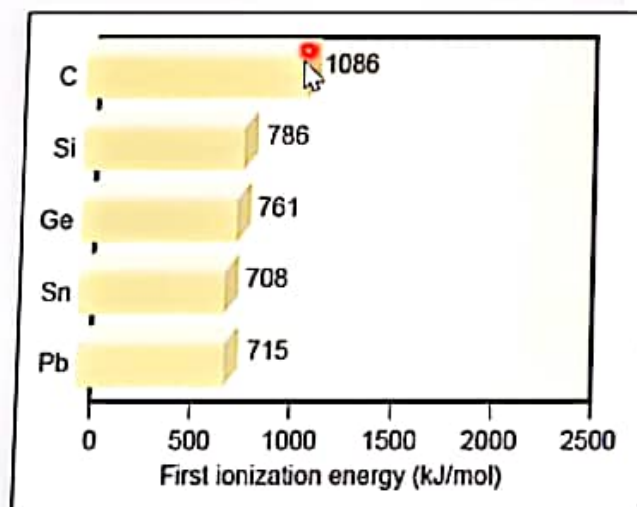
- Both of these elements exhibit the electrical properties of a semiconductor
- Both B and Si and their mineral oxoanions (i.e; borates and silicates) occur in extended covalent networks.
- Both boric acid $[B(OH)_3]$ and silicic acid $[Si(OH)_4]$ are weakly acidic solids that held together by H bonding.
- Their hydrides (boranes silanes) are both flammable, low-melting compounds that act as reducing agents.

Atomic Properties

Atomic radius (pm)		Ionic radius (pm)
C 77		
Si 118		
Ge 122		
Sn 140		Sn ²⁺ 118
Pb 146		Pb ²⁺ 119

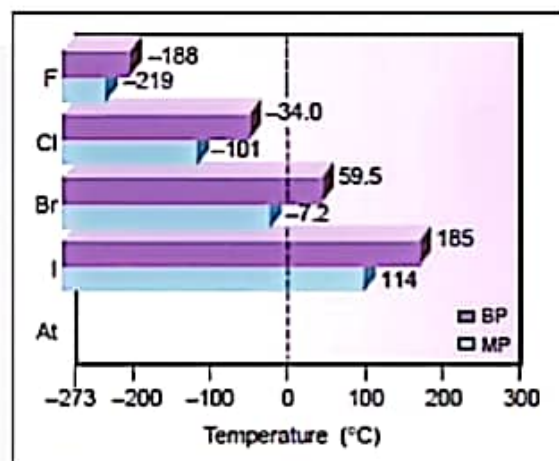
Group electron configuration is ns^2np^2 . Down the group, the number of oxidation states decreases, and the lower (+2) state becomes more common.

Down the group, size increases. Because transition and inner transition elements intervene, IE and EN do not decrease smoothly.

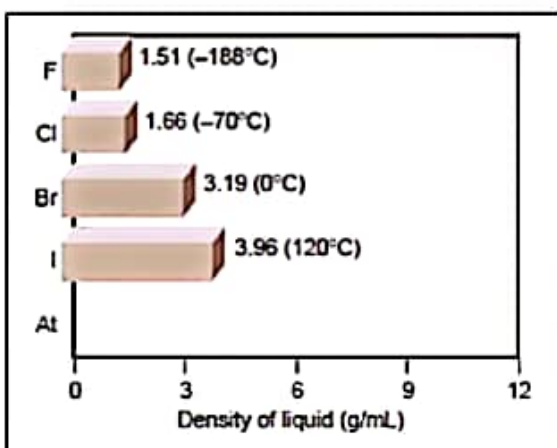


Physical Properties

Down the group, melting and boiling points increase smoothly as a result of stronger dispersion forces between heavier molecules.



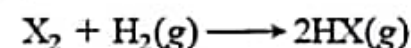
The densities of the elements as liquids (at given T) increase steadily with molar mass.



Important Reactions

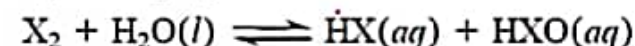
Oxidizing strength and aqueous redox chemistry are shown in reactions 1 and 2, and industrial processes involving fluorine in reactions 3 and 4.

1. The halogens (X_2) oxidize many metals and nonmetals. The reaction with hydrogen, although not used commercially for HX production (except for high-purity HCl), is characteristic of these strong oxidizing agents:



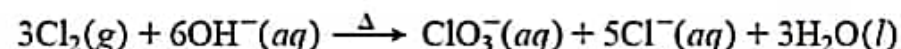
Handwritten note: HCl

2. The halogens disproportionate in water:

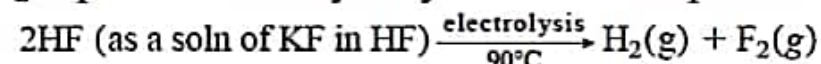


($X = Cl, Br, I$)

In aqueous base, the reaction goes to completion to form hypohalites (*see text*) and, at higher temperatures, halates; for example:

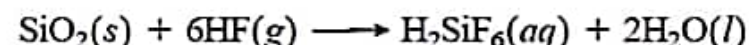


3. F_2 is produced electrolytically at moderate temperature:



A major use of F_2 is in the preparation of UF_6 for nuclear fuel.

4. Glass (amorphous silica) is etched with HF :



Important Compounds

1. Fluorspar (fluorite), CaF_2 . Widely distributed mineral used as a flux in steelmaking and in the production of HF (see photo).

2. Hydrogen fluoride, HF . Colorless, extremely toxic gas used to make F_2 , organic fluorine compounds, and polymers. Also used in aluminum manufacture and in glass etching (margin note, p. 614).

3. Hydrogen chloride, HCl . Extremely water-soluble gas that forms hydrochloric acid, which occurs naturally in the stomach fluids of mammals (humans produce 1.5 L of 0.1 M HCl daily) and in volcanic gases (from reaction of H_2O and sea salt). Made by reaction of NaCl and H_2SO_4 and as a byproduct of plastics (PVC) production. Used in the "pickling" of steel (removal of adhering oxides) and in the production of syrups, rayon, and plastic.

4. Sodium hypochlorite, NaClO , and calcium hypochlorite, $\text{Ca}(\text{ClO})_2$. Oxidizing agents used to bleach wood pulp and textiles and to disinfect swimming pools, foods, and sewage (also used to disinfect the Apollo 11 on return from the Moon). Household bleach is 5.25% NaClO by mass in water.

5. Ammonium perchlorate, NH_4ClO_4 . Strong oxidizing agent used in the space shuttle program (see photo).



Fluorite

6. Potassium iodide, KI . Most common soluble iodide. Added to table salt to prevent thyroid disease (goiter). Used in chemical analysis because it is easily oxidized to I_2 , which provides a colored end point.

7. Polychlorinated biphenyls, PCBs. Mixture of chlorinated organic compounds used as nonflammable insulating liquids in electrical transformers. Production discontinued because of persistence in the environment, where they become concentrated in the tissues of fish, birds, and mammals, and cause reproductive disturbances and possibly cancer.



Space shuttle

I. Electronic Configurations.

F (Z = 9) - (2 s²p⁵)

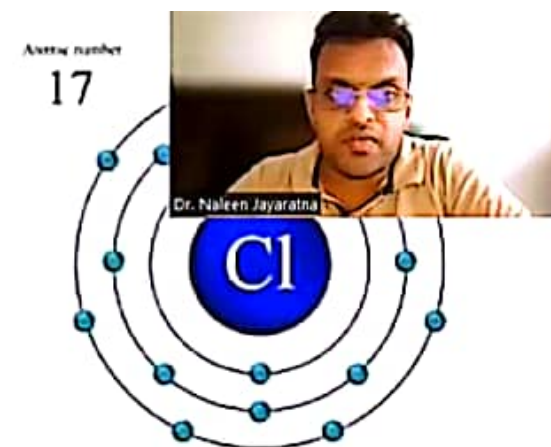
Cl (Z = 17) - (3 s²p⁵)

Br (Z = 35) - (4 s²p⁵)

I (Z = 53) - (5 s²p⁵)

At (Z = 85) - (6 s²p⁵)

- All the halogens have seven electrons (ns²p⁵) in their outer most shell.
- Two electrons are in ns-orbital and the remaining five electrons are in np-orbitals and are distributed as np²x, np²y, np¹z in accordance with Hund's rule.
- The shell previous to the outermost shell contains in F the group s² (saturated), in Cl s²p⁶ (saturated), in Br s²p⁶d¹⁰ (saturated) while in I and At s²p⁶d¹⁰ (unsaturated).
- This shows why F differs from Cl and these two from the remaining elements. Since the additional electron in these elements enters the p orbital of the outermost shell, these are called p-block elements of the long form of the periodic table.





Similarities in Physical Properties

1. Occurrence.

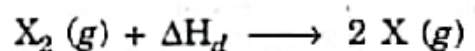
These elements do not occur in the free state. But in the combined state they occur fairly abundantly in nature.

2. Diatomic Nature.

- Exist as covalent diatomic molecules (X_2 molecule) in the gaseous, liquid and crystalline states.
- These X_2 molecules are held together by weak Van der Waals forces which explain the volatile nature of these elements.
- At high temperatures, X_2 molecules dissociate into X atoms.
- The stability of diatomic molecules towards heat decreases from F_2 to I_2 .

3. Heat of Dissociation of X_2 Molecules.

- Heat of dissociation (H_d) of X_2 molecule is the amount of heat required to dissociate X_2 molecule in the gaseous state into X gaseous atoms or to break X—X bond in X_2 molecule.



Obviously, ΔH_d is equal to X—X bond energy. Excepting for F_2 these energies decrease from Cl_2 to I_2 .

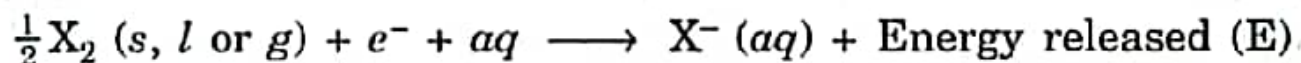
-The lower value of heat of dissociation of F_2 is due to the reasons;

(i) Due to very small F—F bond length, repulsion between the non-bonding electrons is very large as compared to other X—X bond lengths (X = Cl, Br, I). This makes the F atoms in F_2 molecule repel each other and helps the dissociation of F_2 molecule into F atoms.

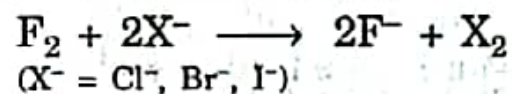
(ii) X—X bond in Cl_2 , Br_2 and I_2 molecules is stronger than F—F bond in F_2 molecule. This is due to the possibility of the existence of multiple bonding in X—X bond involving d-orbitals.



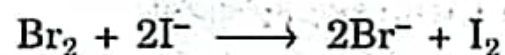
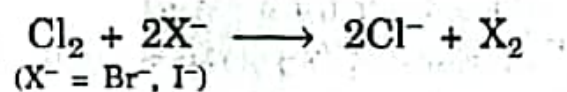
4. Oxidizing Property. A substance that has a tendency to accept one or more electrons is said to show oxidizing property. Halogens, due to high electron affinity values, have a great tendency to accept an electron and hence act as oxidizing agents.



- Since the values of E are decreasing from F_2 to I_2 the oxidizing power of halogens is also decreasing in the same direction, i.e. the oxidizing power of halogens is in the order: F_2 (strongest oxidizing agent) $>$ $Cl_2 > Br_2 > I_2$ (weakest oxidizing agent).
- Since F_2 is the strongest oxidizing agent in the series, it will oxidize other halide ions to halogens in solution or when dry, i.e. F_2 displaces other halogens from their corresponding halides.



- Similarly Cl_2 will displace Br^- and I^- ions from their solutions and Br_2 will displace I^- ions from their solutions.





5. Formation of Clathrates.

Water solutions of the halogens on cooling give the crystals of their clathrates of $X_2 \cdot xH_2O$ type. Examples are : $Cl_2 \cdot 7.3 H_2O$, $Br_2 \cdot 10H_2O$, $I_2 \cdot xH_2O$ etc.

6. Electronegativity.

Halogens have large values of electronegativity. These values decrease from F to I in the group. Large electronegativity values of halogen atoms indicate that X atoms have a strong tendency to form X^- ions.

7. Solubility.

- The halogens are slightly soluble in water and their solubility decreases from Cl to I.
- F cannot be dissolved in H_2O , since it decomposes H_2O liberating O_2 and some O_3 .
- The halogens dissolve much better than H_2O in many organic solvents like carbon disulphide, chloroform, ether etc.
- In organic solvents, iodine gives violet colour, bromine yields brown colour and chlorine produces yellow colour.
- It is assumed that in organic solvents halogens exist as free molecule as in the gas phase. In nucleophilic (electron donating) polar solvent like alcohols, ketones the halogens give coloured solutions whose colours are due to the formation of solvent -- halogen type complexes. These complexes are charge-transfer compounds.

Similarities in Chemical Properties

1. Reactivity.

- All the halogens are the most reactive elements as a family.

The main reasons for their extreme reactivity are :

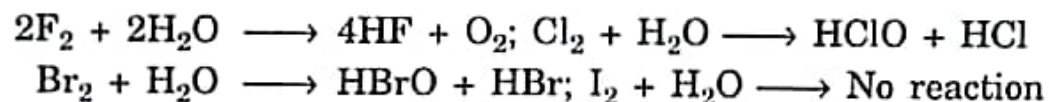
(a) Low dissociation energies of X_2 molecules.

For example: $O_2 = 495.08$, $F_2 = 157.77$, $Cl_{12} = 237.38$, $Br_2 = 189.99$ in kJ/mol.

(b) High electron affinity values of halogen atoms - halogens have a strong tendency to add on electron to form X ion.

The reactivity of halogen atoms decreases down the group from F to I.

- For example:
- fluorine decomposes H_2O vigorously even in dark,
 - chlorine decomposes H_2O in sunlight
 - bromine also decomposes H_2O in sunlight but very slowly
 - iodine does not decompose water at all



2. Formation of Hydracids.

- All the elements of this group combine with hydrogen and form hydrogen halides (HX) which are called hydracids or halogen acids.
- The affinity of these elements for hydrogen decreases from fluorine to iodine.
 - F_2 combines with H_2 explosively and even in the dark
 - Cl_2 reacts with H_2 in diffused sunlight,
 - Br_2 reacts with H_2 only on heating above $500^\circ C$
 - I_2 reacts with H_2 only on heating and in presence of Pt as a catalyst.
- Methods of formation of Hydracids are such as (i) direct combination of the elements (ii) heating the metal halides with conc. H_2SO_4 (iii) hydrolysis of non-metallic halides.
- *Some important properties of hydracids are given below.*

(i) Acidic properties.

All the halogen acids in the gaseous state are essentially covalent but in the aqueous solution they ionize to give the solvated proton (H_3O^+) and hence act as acids.



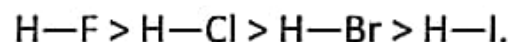
HF ionizes only slightly while HCl, HBr and HI ionize completely. Hence HF is the weakest acid and the strength of these acids increases from HF to HI,

i.e. HF (weakest acid) < HCl < HBr < HI (strongest acid).

The weakest acidic nature of HF is due to the fact that the dissociation energy of H—F bond in HF molecule is the highest (= 564.97 KJ.mol⁻¹) and hence this molecule has least tendency to split up into H^+ and F^- ions in aqueous solution

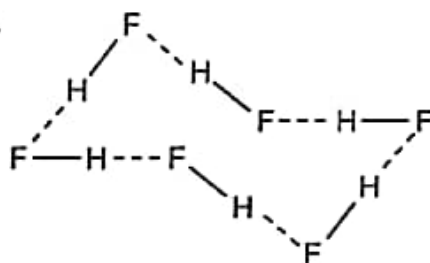
(ii) Covalent character.

- The bond between hydrogen and the halogen in these hydrides is largely covalent.
- The bonds do, however, have small amount of polar character which decreases with the decrease of electronegativity of the halogen atom. Thus, the polar character of H—X bond in HX molecules decreases in the order:



(iii) Physical properties.

- Anhydrous HF is a liquid at ordinary temperature whereas other HX (X = Cl, Br, I) are colourless gases.
- Excepting HF, the melting and boiling points of these hydrides increase with the increase of the atomic weight of the halogen atom.
- The abnormally high melting point of HF is explained by saying that, due to H-bonding, HF molecule gets associated to form (HF)₆ which is an aggregated molecule.



- The values of heat of formation increase with the increase of atomic weight of halogen atom. Thus the quantity of energy released during the formation of HX molecules decreases from HF to HI.

(iv) **Stability and reducing property.**

- As we go from HF to HI, the stability of these acids decreases from HF to HI. (HF > HCl > HBr > HI).
- Thus HF and HCl are stable upto 1200°C while HBr dissociates slightly (I) and HI dissociates at 440°C to 20%. The decrease in the stability of HX molecules from HF to HI implies that the reducing properties of these acids increase from HF to HI,

i.e. HF < HCl < HBr < HI.

- The above order shows that HF has no appreciable reducing properties while HI is the strongest reducing agent.
- Even dilute solution of HI, on exposure to the air, slowly acquires a brown colour due to the liberation of I₂.



3. Formation of Binary Halogen-oxygen Compounds.

- Halogens do not combine directly with O₂. However, binary halogen-oxygen compounds can be prepared by indirect methods.
- Eg: OF₂, O₂F₂, Cl₂O, ClO₂, Br₂O, BrO₂, I₂O₄.
- F being more electronegative than oxygen, OF₂ and O₂F₂ are called fluorides of oxygen and not oxides of fluorine.

4. Formation of Oxy-acids.

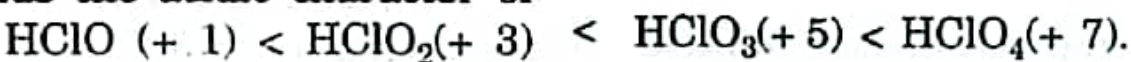
- Excepting F, all other halogens form oxy-acids.
- These are mainly of four types

- HXO (e.g. HClO, HBrO, HIO)
- HXO₂ (e.g. HClO₂)
- HXO₃ (e.g. HClO₃, HBrO₃, HIO₃)
- HXO₄ (e.g. HClO₄, HBrO₄, HIO₄)

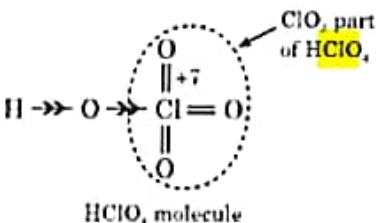
Some of their properties are given below

(i) Acidic character.

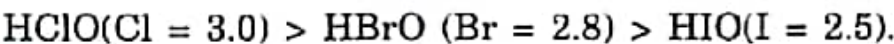
(a) In the series of oxy-acids of the same halogen in different oxidation number the acidic character of such acids increases with the increase of the oxidation number of the halogen. Thus the acidic character of HClO, HClO₂, HClO₃, HClO₄ acids is in the order.



The above order of acidic character of the acids is explained by finding out the relative order of the basicity of the conjugate bases (viz. ClO⁻, ClO₂⁻, ClO₃⁻ and ClO₄⁻) of these acids.



(b) The acidic character of the oxy-acids of different halogens in the same oxidation state decreases, in general, with decrease of the electronegativity of the halogen atom. Thus the acidic character of HClO, HBrO and HIO acids is in the following order.



Electronegativity of halogen atoms is given in bracket.

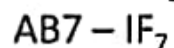
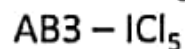
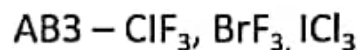
- As the atomic number of the halogen atom increases, its electronegativity decreases and size increases; As a result of this, the tendency of these acids to lose a proton to water decreases. This makes the acid a weaker acid.

(ii) Oxidizing power.

- As the oxidation number of the halogen atom increases, X—O bond becomes more covalent, resulting in a decrease of the oxidizing power of the acid.
- For example under usual laboratory conditions, the oxidizing action of HClO, HClO₂, HClO₃, and HClO₄ acids is in the order HClO(+ 1) > HClO₂(+ 3) > HClO₃(+ 5) > HClO₄(+ 7).

5. Formation of Interhalogen Compounds.

- Two different halogen atoms, due to the difference in their electronegativity values, combine with one another to form a number of covalent compounds which are called interhalogen compounds.



Anomalous Behaviour of Fluorine

- Fluorine differs from the other members of its own subgroup due to
 - (a) its small size
 - (b) its high electronegativity
 - (c) non-availability of d-orbitals in its valence shell.

The main differences are :

- (i) Due to its maximum electronegativity, it shows only a negative oxidation state of -1. It does not show any positive oxidation state. The other members show negative as well as positive oxidation states of + 1, + 3, + 5 and + 7.
- (ii) Fluorine is the most reactive element among halogens. This is due to the minimum value of F—F bond dissociation energy.
- (iii) On account of high electronegativity of fluorine atom hydrogen bonding occurs in HF molecule while it is absent in HCl, HBr and HI molecules. The presence of hydrogen bonding in HF explains why
 - (a) HCl, HBr and HI are gaseous while HF is a liquid with an abnormally high boiling point
 - (b) HF forms a number of compounds containing HF_2^- ion are not given by other HX molecules.
- (iv) HF ionizes only slightly in aqueous solution and is, therefore a weak acid. HCl, HBr and HI, on the other hand, ionize almost completely and are thus strong acids.

- (v) Fluoride of silver is soluble while other halides of silver are insoluble. Similarly fluoride of calcium is insoluble and other halides of calcium are soluble.
- (vi) Fluorides are ionic while other halides have frequently molecular lattices.
- (vii) F ion has a great tendency to give complex ions like $[\text{AlF}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ etc. The remaining halide ions show this tendency to a much lesser extent.
- (viii) Fluorine, being the most electronegative, gives SF_6 while other members do not form hexahalides with sulphur.
- (x) Because of the absence of d-orbitals in its valence shell, fluorine does not combine with F ions to give polyhalide ion, F_3^- while other members of the group give such ions (e.g. Cl_3^- , Br_3^- , I_3^- , I_5^- etc.) because they contain d-orbitals.

Comparison of Fluorine and Oxygen.

Fluorine resembles oxygen on following factors :

- (a) The atomic and ionic radii of both the elements are very similar
- (b) Both are highly electronegative.

- **Similarities.**

- (i) Both of them exist in gaseous state at ordinary temperature
- (ii) Both react with metals (e.g. Na) and non- metals (e.g., H, C, Si, S etc.) to give fluorides and oxides which resemble each other in solubility, volatility, fusibility etc., e.g.

HF and H₂O resemble each other in the following respects

- (a) Both possess associated molecules, (H₂O) and (HF) respectively due to hydrogen bonding
- (b) The boiling points of both are abnormally high
- (c) HF like H₂O is a weak acid and both are ionizing solvents.
- (d) Heats of formation of HF and H₂O are close to each other.

(iii) Certain elements like sulphur exhibit higher valencies in both fluorides and oxides, e.g. the valency of sulphur in SF₆, and SO₃ is 6.

Dissimilarities.

However these elements differ from each other in the following respects :

- (i) F is always monovalent while oxygen is bivalent
- (ii) F is more reactive than oxygen.